

ECE8803-AI4

Module: Fault Detection and Classification

Supplementary Lecture: Principal Component Analysis

Course website: <https://aikhan123.github.io/khan-lab-website/ai-semiconductors-course.html>

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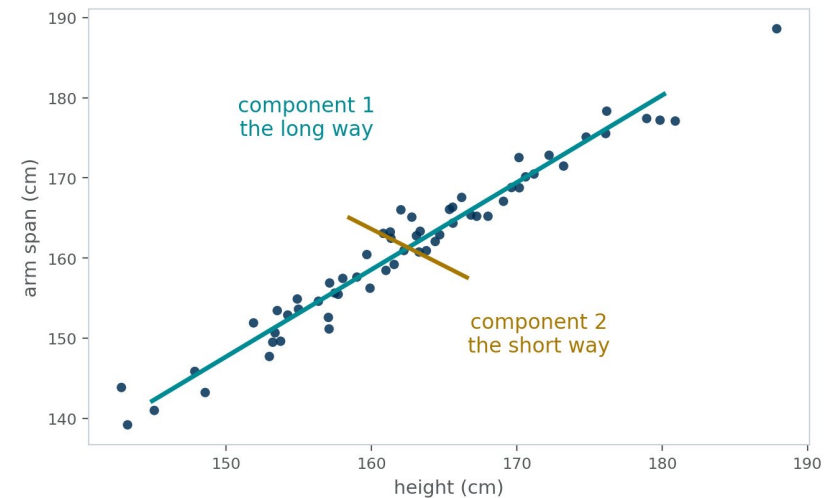
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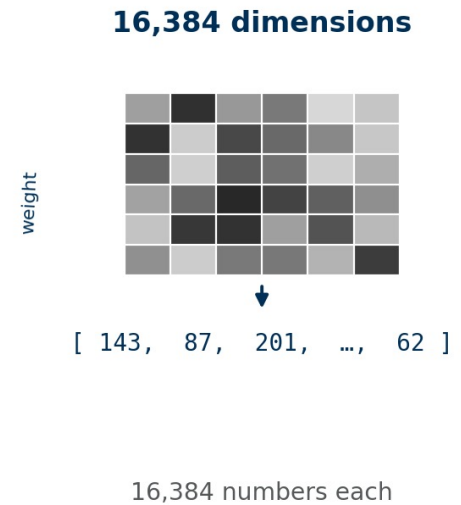
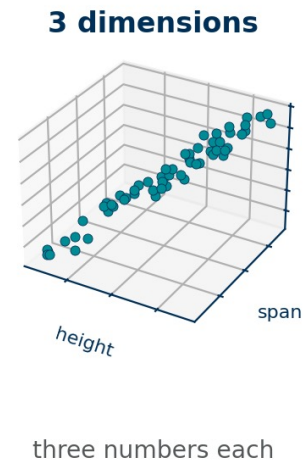
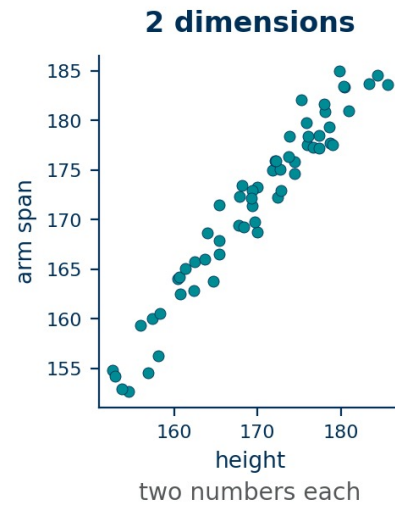
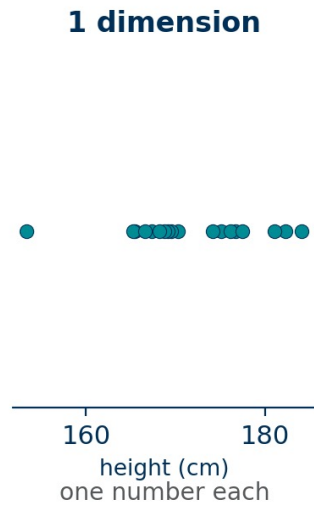
Before faces: two numbers per person

- Measure everyone's height and arm span. The dots form a stretched-out sausage.
- If you could keep only one number per person, you would not keep height and you would not keep arm span. You would keep the position along the sausage.
- That direction is component 1. The thin direction across the sausage barely varies, so throwing it away costs almost nothing.
- Two measurements turned out to be one thing wearing two hats: bigness.



What a dimension is

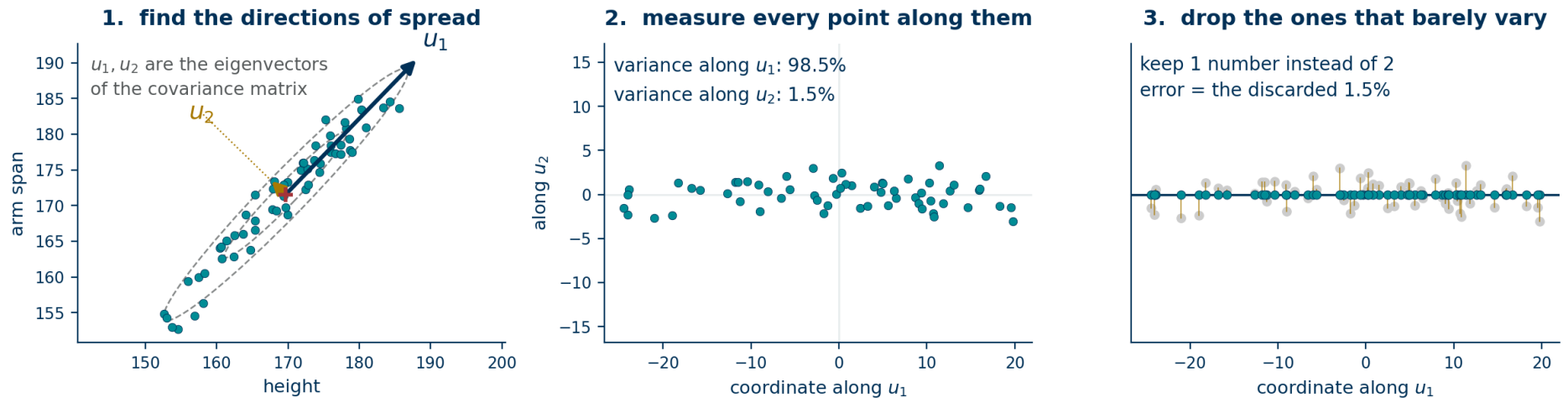
A dimension is one number in the list that describes an object. Measure height and you have one dimension; add arm span and you have two; add weight and you have three. A 128×128 photograph is a list of 16,384 numbers, so it lives in 16,384 dimensions.



Nobody can draw sixteen thousand axes, and nothing about the idea needs drawing. A point is still a list of numbers, two points are still close or far apart, and a direction is still a direction.

PCA finds the eigenvectors of the data

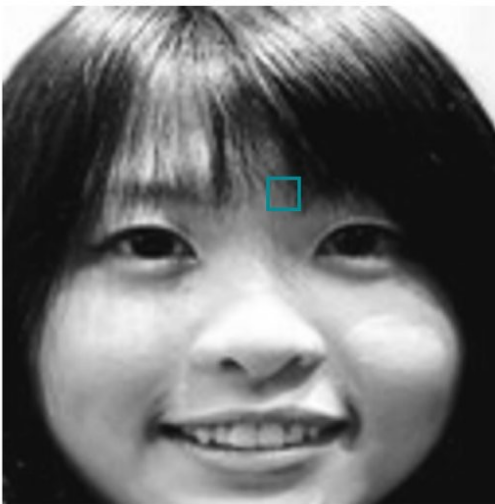
Take the cloud of measurements, work out how they vary together — that is the covariance matrix — and find its eigenvectors. Each eigenvector is a direction through the cloud, and its eigenvalue is how much the data spreads along it.



Keeping the directions with large eigenvalues and discarding the rest is dimensionality reduction. The rest of this lecture is these three steps, run in 16,384 dimensions instead of two.

A photo is already a list of numbers

Each pixel holds one brightness value from 0 (black) to 255 (white). A 128 x 128 photo is 16,384 of them, in a fixed order. Nothing has been done to the picture yet — this is just what an image is.



one 128 x 128 photo = 16,384 brightness numbers



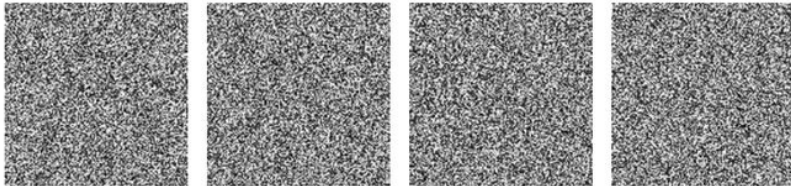
85	74	59	64	63	39	28	25
74	88	58	65	61	38	32	28
80	97	73	61	65	55	41	20
91	91	84	78	64	52	56	38
96	91	88	85	63	52	63	46
105	90	88	89	72	61	73	40
111	90	93	94	80	66	78	45
109	96	100	96	92	80	78	58

zoom in far enough and it was numbers all along

Faces sit on a thin sheet inside a huge space

If a photo is 16,384 numbers, then a photo is a single point in 16,384-dimensional space. Now pick 16,384 numbers at random: you get static, every time. Faces are a vanishingly small, thin sheet inside that enormous space — and PCA's job is to find the sheet.

16,384 random numbers



almost every point in the space looks like this

16,384 numbers that make a face



faces are a vanishingly thin sheet inside it

Thirty faces, all lined up the same way



Alignment matters: eyes in roughly the same place in every photo. Feed PCA unaligned pictures and it wastes its components describing wobble.

Photographs: Flickr-Faces-HQ (FFHQ), NVIDIA — Flickr portraits under permissive licences, cropped to the face and converted to grey. Dataset licence CC BY-NC-SA 4.0: fine for teaching, not for commercial use.

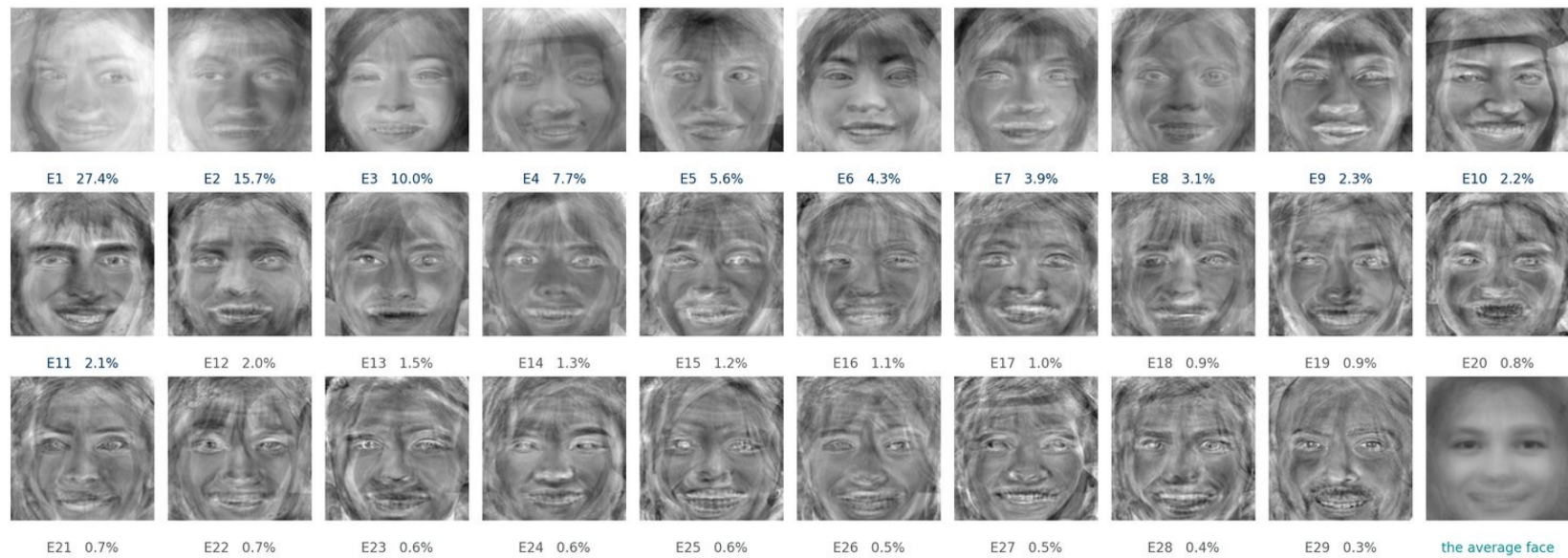
Average all thirty



- Add up all thirty faces pixel by pixel and divide by thirty.
- This is the mean face. It belongs to nobody in the class.
- It is our starting point — the guess we would make if we knew nothing at all about a person.
- Everything from here on describes how someone differs from this face.

And here are the eigenfaces

All twenty-nine of them, ranked by how much of the variation each one explains, with the average face again at the end for comparison. Not one of them is a person. Mid-grey means no change; lighter areas add brightness, darker areas take it away.



One eigenface

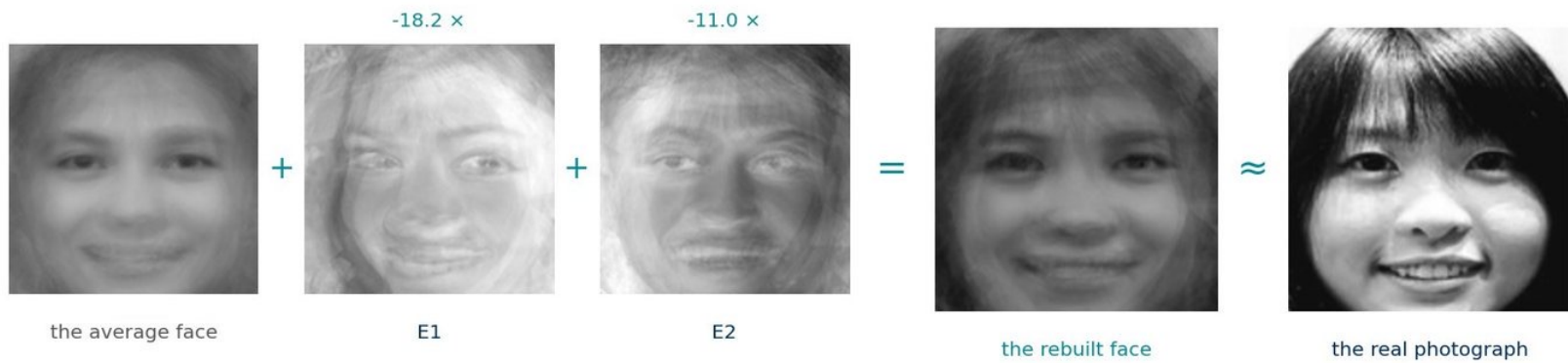
The strongest pattern, scaled by -18.2 and added to the average. Exposure and face shape are roughly right. The person is not.



average pixel still 22.5% off

Two eigenfaces

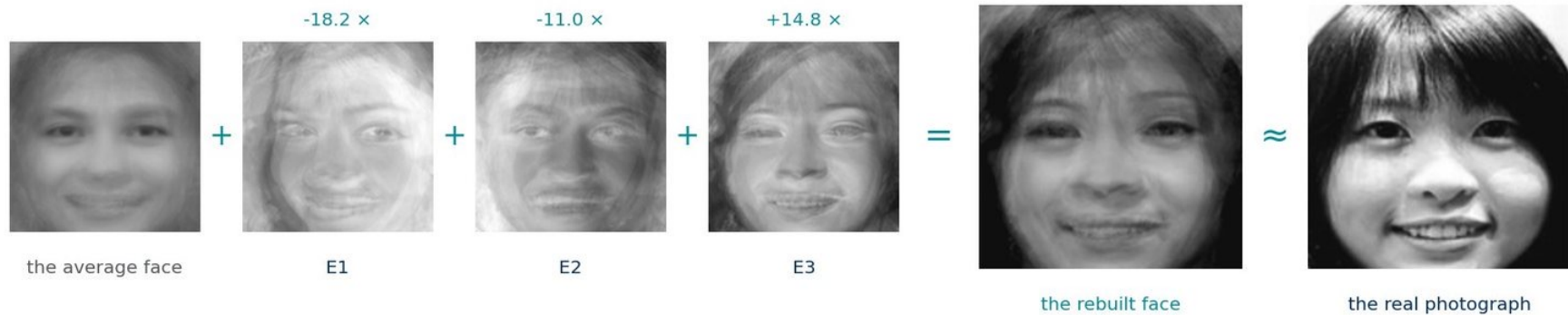
The second darkens the frame around her. Still a generic face.



average pixel still 21.7% off

Three eigenfaces

The third begins putting hair on her head. Now it is somebody rather than anybody.



average pixel still 17.7% off

Four eigenfaces

Her largest amount of all, +22.4 of E4 — the dark fringe and the shape of her face arrive in a single step. The error falls from 17.7% to 11.7%.

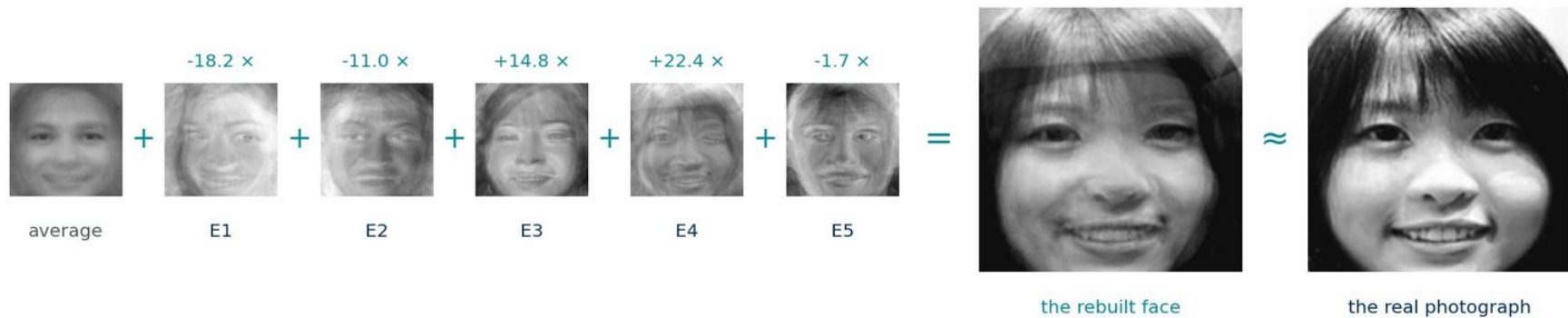


average pixel still 11.7% off

78% of her captured

Five eigenfaces

Almost nothing changes. Her fifth amount is only -1.7 : eigenface five happens to describe something she does not have. The ranking is global, not personal.



average pixel still 11.6% off

78% of her captured

Six eigenfaces

Six terms no longer fit on one line, so the eigenfaces stack into a block from here. This one matters: 11.6% down to 7.9%.



E1 to E6 in reading order — the teal number over each one is how much of it this face needs

average pixel still 7.9% off

89% of her captured

Seven eigenfaces

Seven. The last of the big structural work. Compare this frame with the one before — then keep comparing, because the changes are about to stop.



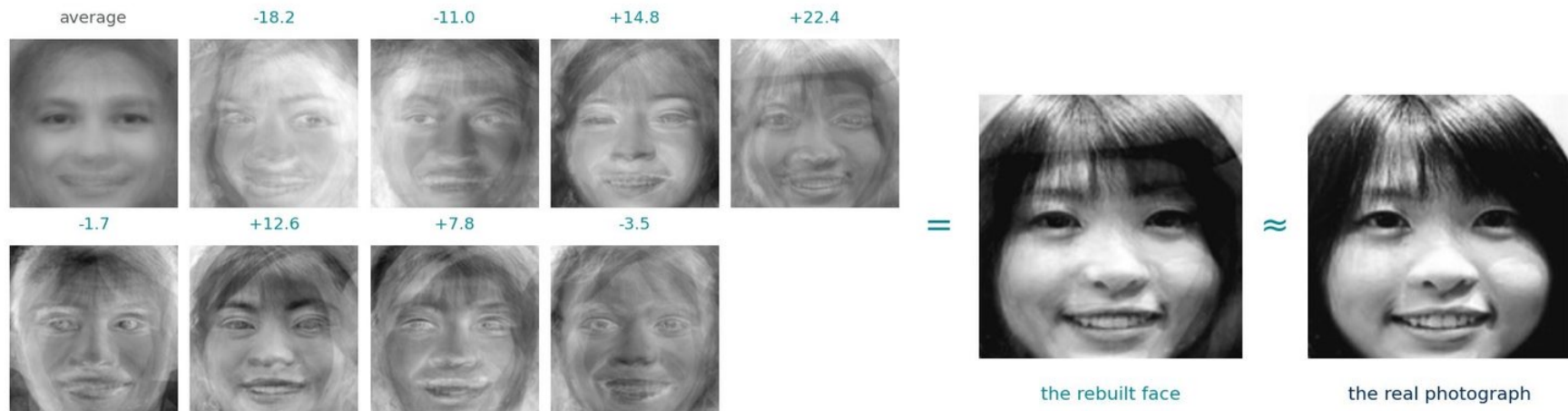
E1 to E7 in reading order — the teal number over each one is how much of it this face needs

average pixel still 6.5% off

93% of her captured

Eight eigenfaces

Eight eigenfaces carry 93% of everything that makes her different from the average. Look hard at this frame: the twenty-one still to come will barely disturb it.



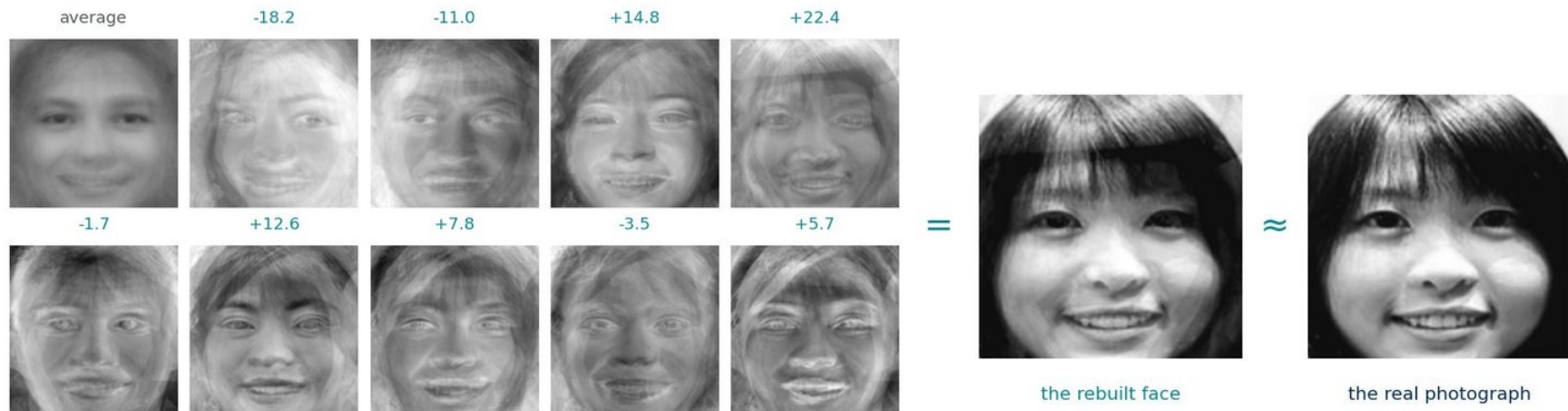
E1 to E8 in reading order — the teal number over each one is how much of it this face needs

average pixel still 6.1% off

93% of her captured

Nine eigenfaces

Nine. From here on you are watching an error percentage fall, not a face change.

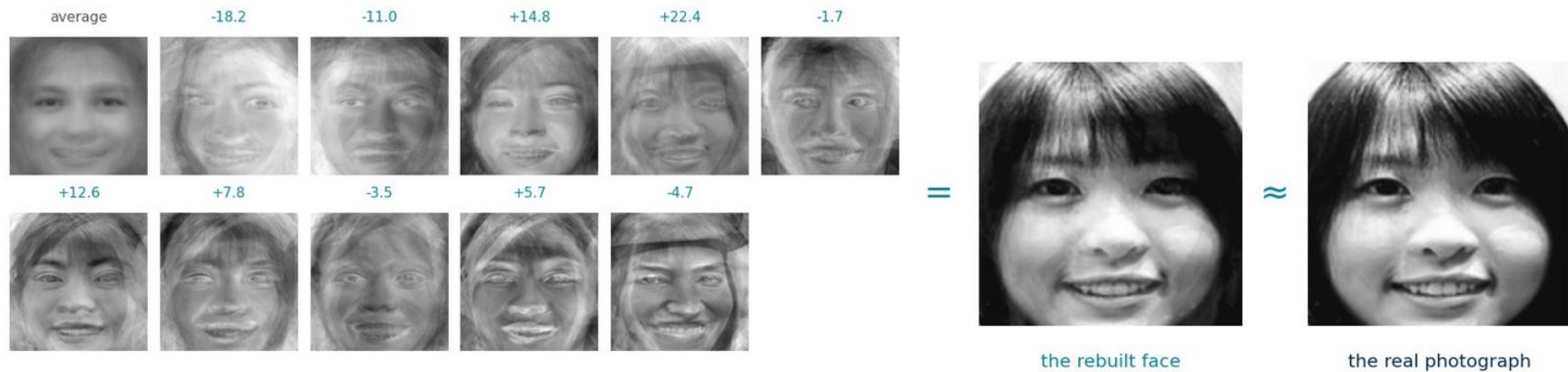


E1 to E9 in reading order — the teal number over each one is how much of it this face needs

average pixel still 5% off

Ten eigenfaces

Ten. 97% of the way there. Flip back to slide 10 and forward again — the difference between five and ten is obvious; the difference between ten and twenty will not be.



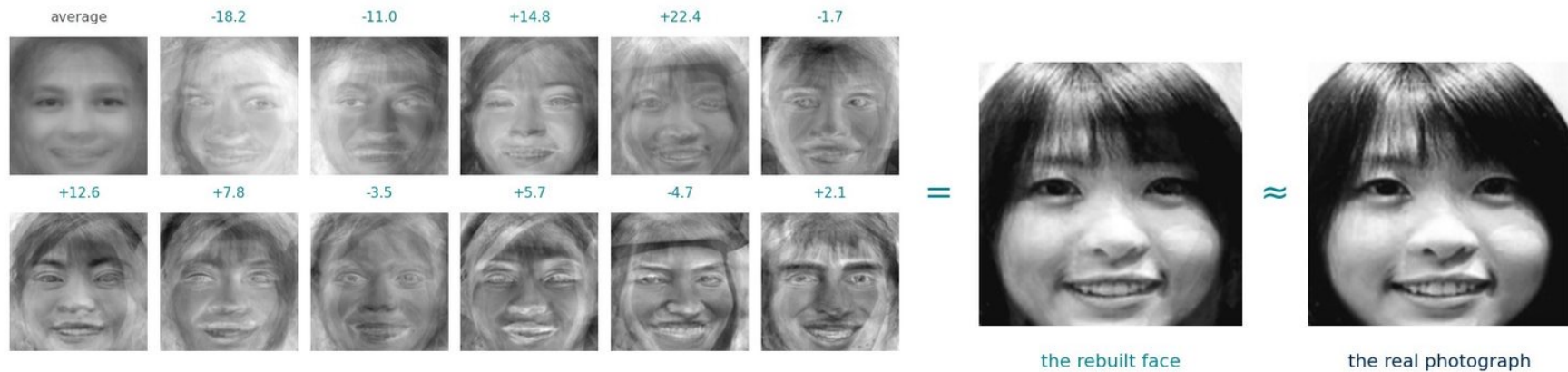
E1 to E10 in reading order — the teal number over each one is how much of it this face needs

average pixel still 4.2% off

97% of her captured

Eleven eigenfaces

Eleven terms. The picture has stopped changing; only the error percentage is still moving.

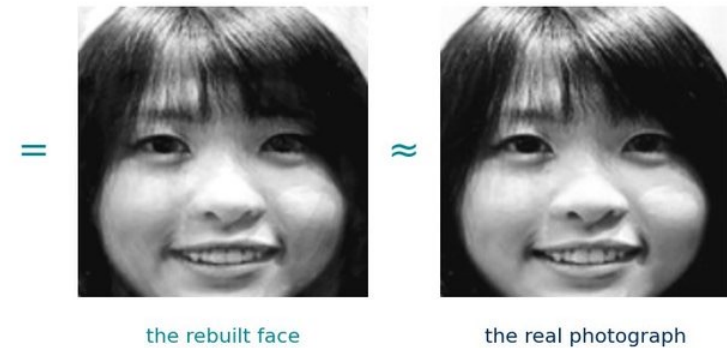


E1 to E11 in reading order — the teal number over each one is how much of it this face needs

average pixel still 4% off

Twelve eigenfaces

Twelve. 98% captured. Each new eigenface is now worth a fraction of a percent.



E1 to E12 in reading order — the teal number over each one is how much of it this face needs

average pixel still 3.1% off

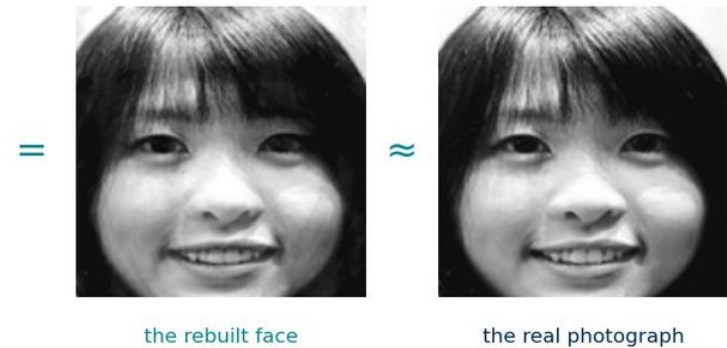
98% of her captured

Thirteen eigenfaces

Thirteen terms. The picture has stopped changing; only the error percentage is still moving.



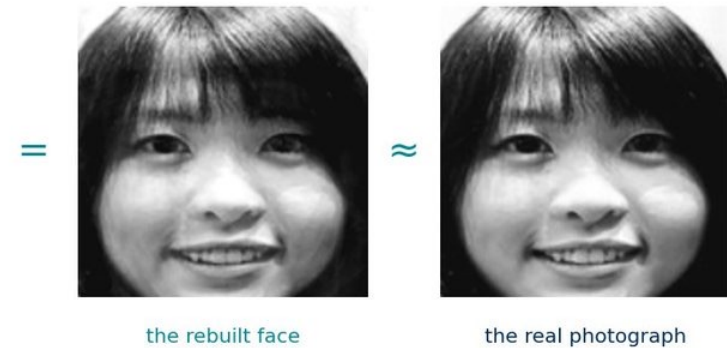
E1 to E13 in reading order — the teal number over each one is how much of it this face needs



average pixel still 3.1% off

Fourteen eigenfaces

Fourteen terms. The picture has stopped changing; only the error percentage is still moving.

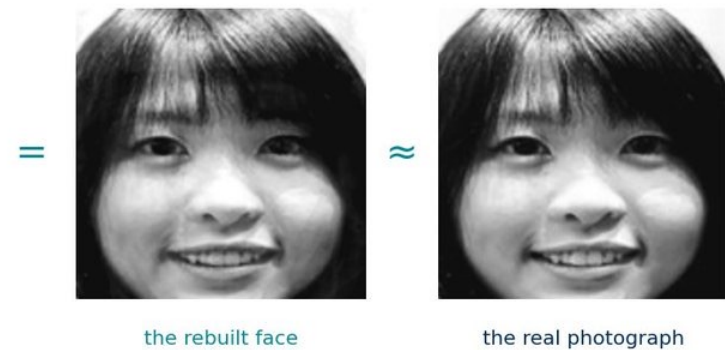
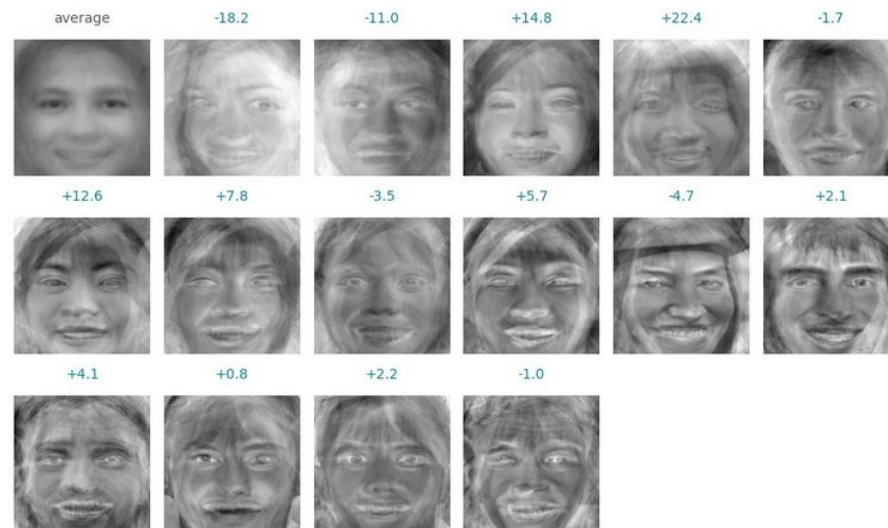


E1 to E14 in reading order — the teal number over each one is how much of it this face needs

average pixel still 2.7% off

Fifteen eigenfaces

Fifteen — half of the available patterns, 99% of her. This is where you would stop if storage cost money.



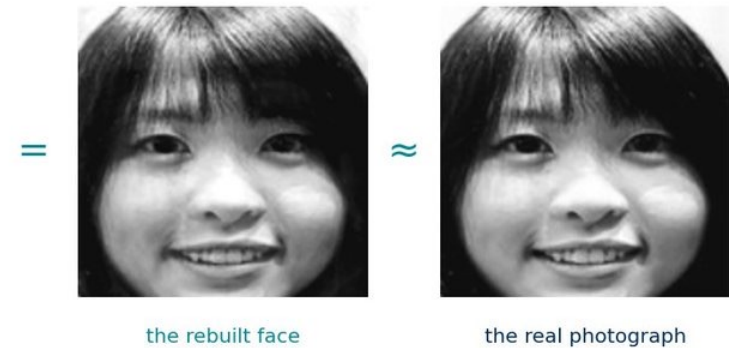
E1 to E15 in reading order — the teal number over each one is how much of it this face needs

average pixel still 2.7% off

99% of her captured

Sixteen eigenfaces

Sixteen terms. The picture has stopped changing; only the error percentage is still moving.



E1 to E16 in reading order — the teal number over each one is how much of it this face needs

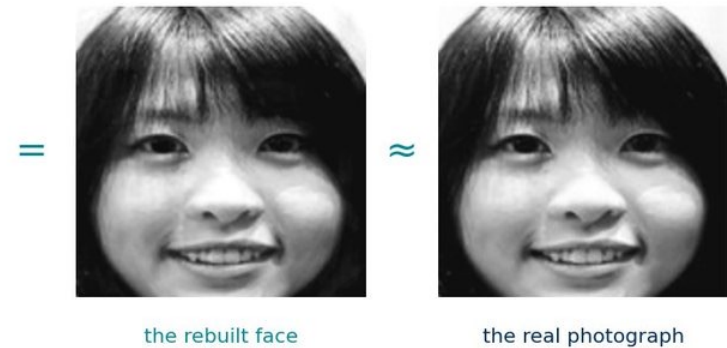
average pixel still 2.4% off

Seventeen eigenfaces

Seventeen terms. The picture has stopped changing; only the error percentage is still moving.



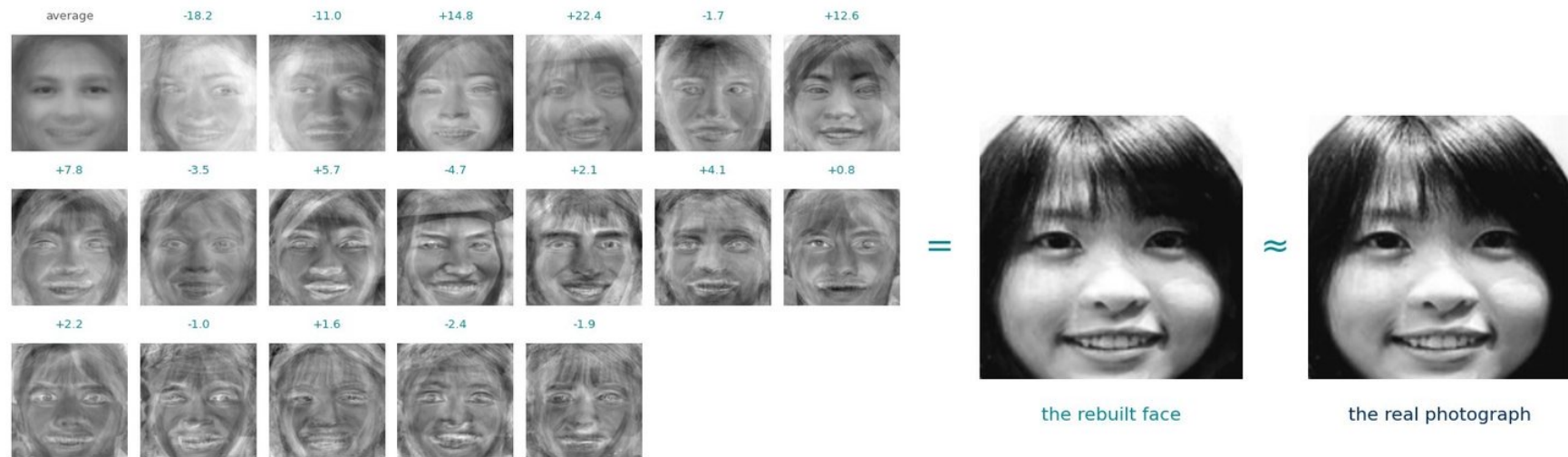
E1 to E17 in reading order — the teal number over each one is how much of it this face needs



average pixel still 1.9% off

Eighteen eigenfaces

Eighteen. If you cannot see any change from the last three slides, that is the lesson, not a failure of the projector.

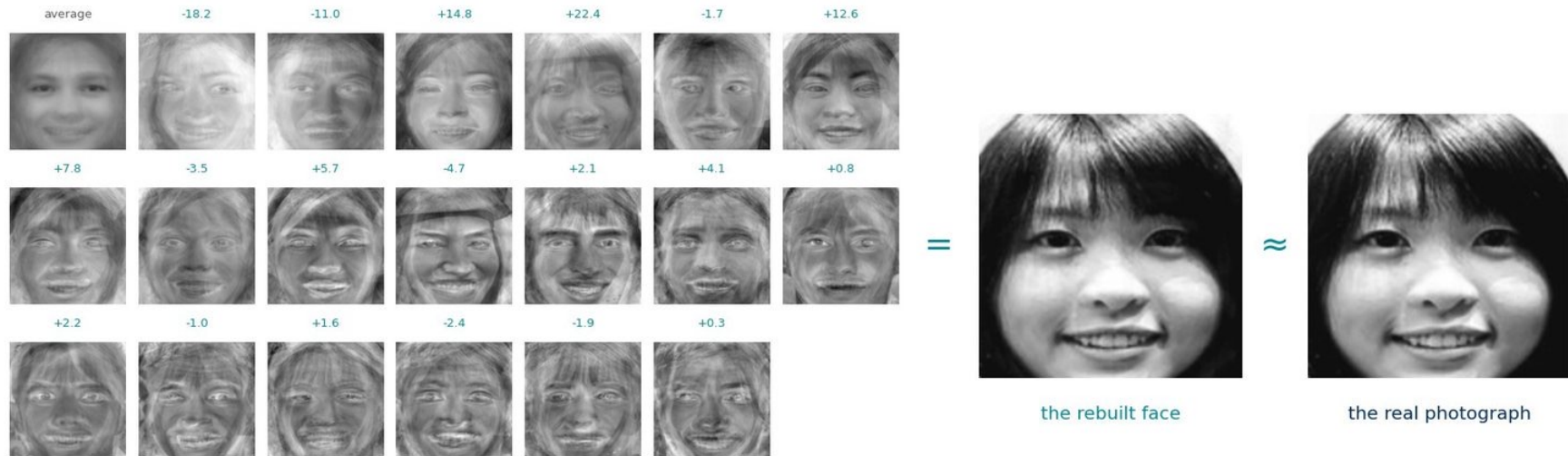


E1 to E18 in reading order — the teal number over each one is how much of it this face needs

average pixel still 1.5% off

Nineteen eigenfaces

Nineteen terms. The picture has stopped changing; only the error percentage is still moving.

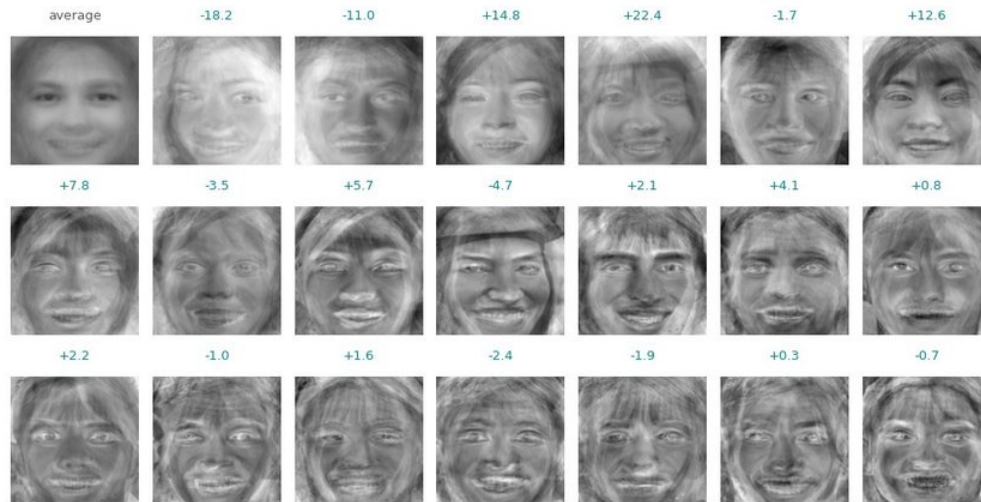


E1 to E19 in reading order — the teal number over each one is how much of it this face needs

average pixel still 1.5% off

Twenty eigenfaces

Twenty. The error is 1.4%: about one part in seventy of the brightness range, spread over 16,384 pixels.



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the rebuilt face

≈



the real photograph

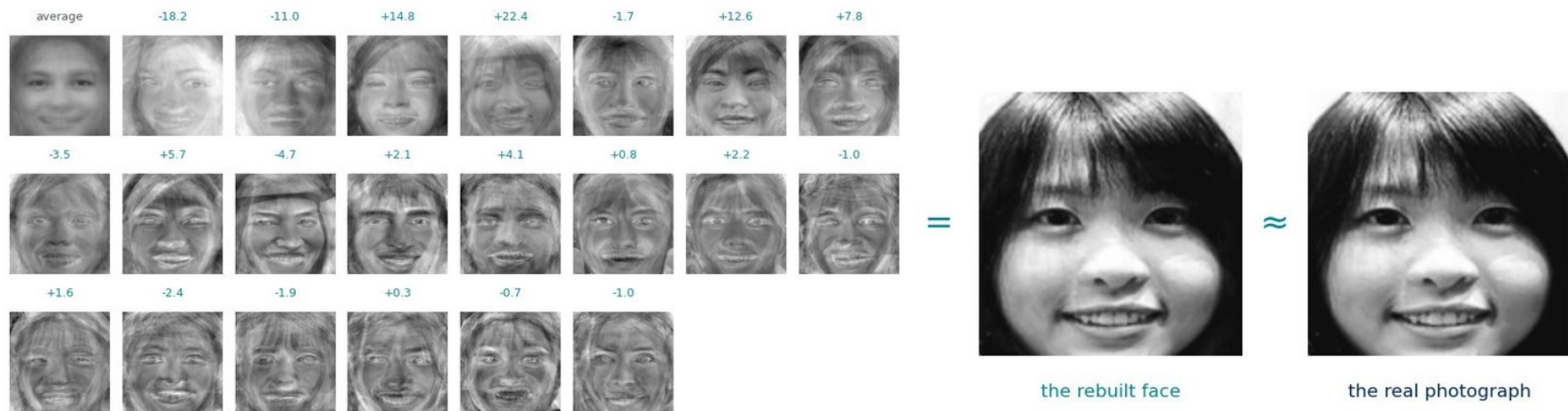
E1 to E20 in reading order — the teal number over each one is how much of it this face needs

average pixel still 1.4% off

99.6% of her captured

Twenty-one eigenfaces

Twenty-one terms. The picture has stopped changing; only the error percentage is still moving.



E1 to E21 in reading order — the teal number over each one is how much of it this face needs

average pixel still 1.3% off

Twenty-two eigenfaces

Twenty-two terms. The picture has stopped changing; only the error percentage is still moving.



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the rebuilt face

≈



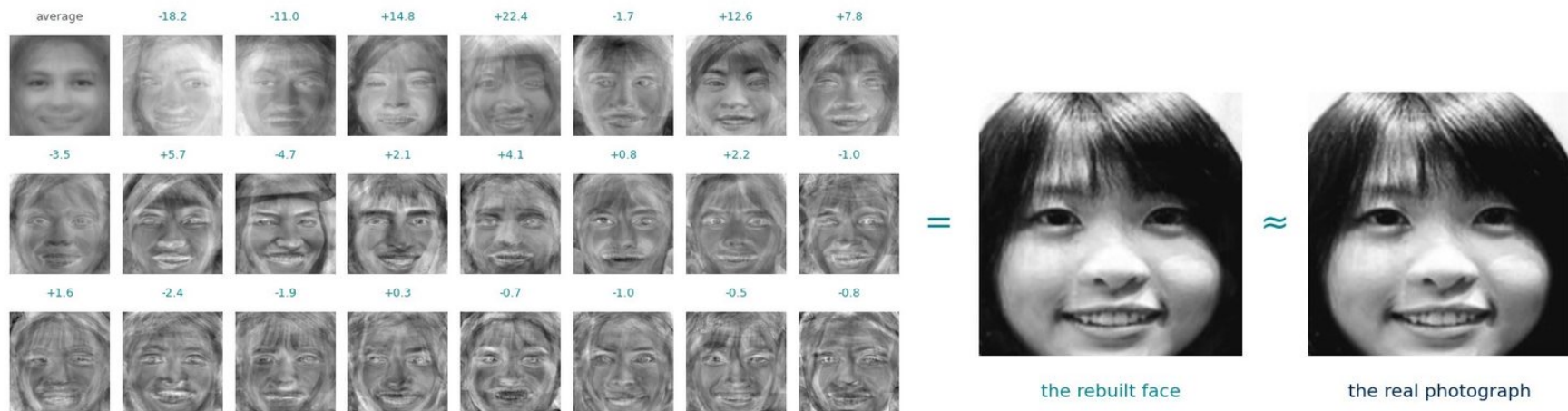
the real photograph

E1 to E22 in reading order — the teal number over each one is how much of it this face needs

average pixel still 1.2% off

Twenty-three eigenfaces

Twenty-three terms. The picture has stopped changing; only the error percentage is still moving.

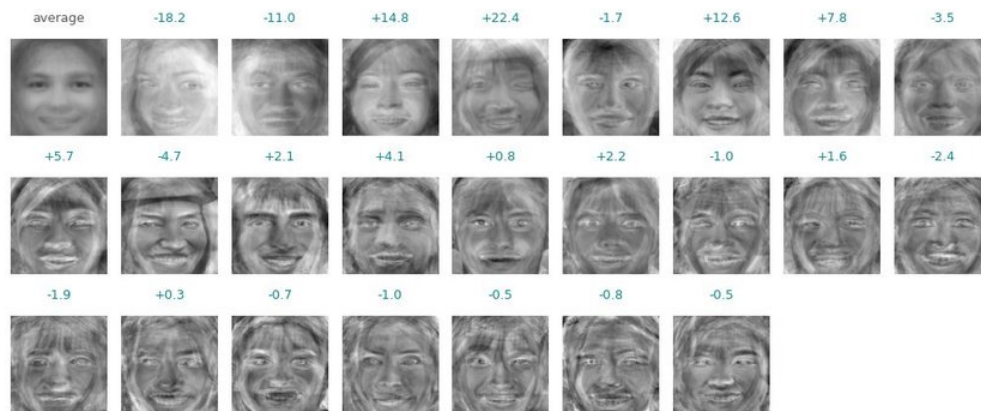


E1 to E23 in reading order — the teal number over each one is how much of it this face needs

average pixel still 1.1% off

Twenty-four eigenfaces

Twenty-four. The remaining eigenfaces describe skin grain and stray hairs.



E1 to E24 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

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the real photograph

average pixel still 1.1% off

Twenty-five eigenfaces

Twenty-five terms. The picture has stopped changing; only the error percentage is still moving.



E1 to E25 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

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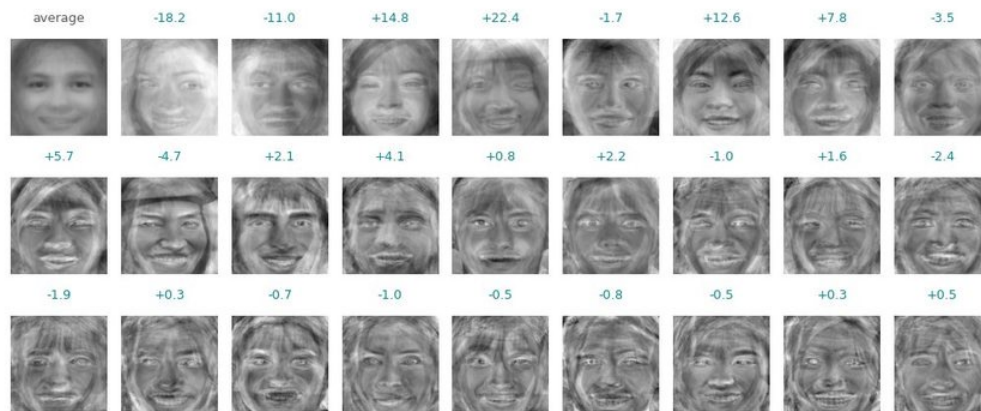
the real photograph

average pixel still 1.1% off

99.8% of her captured

Twenty-six eigenfaces

Twenty-six terms. The picture has stopped changing; only the error percentage is still moving.



E1 to E26 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

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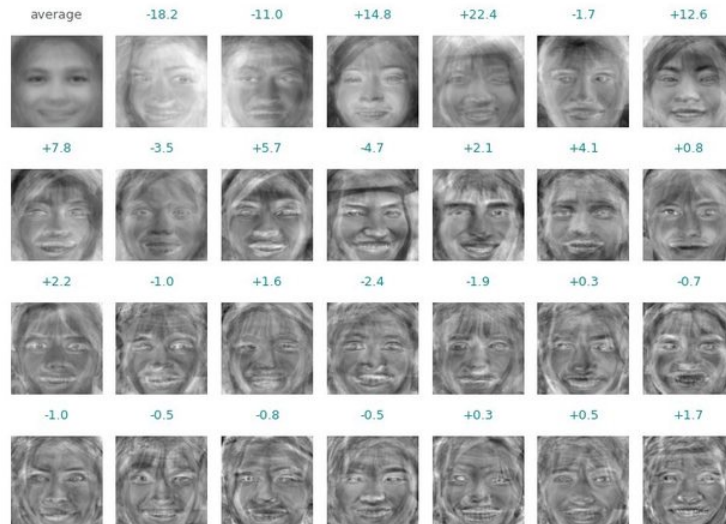


the real photograph

average pixel still 1.1% off

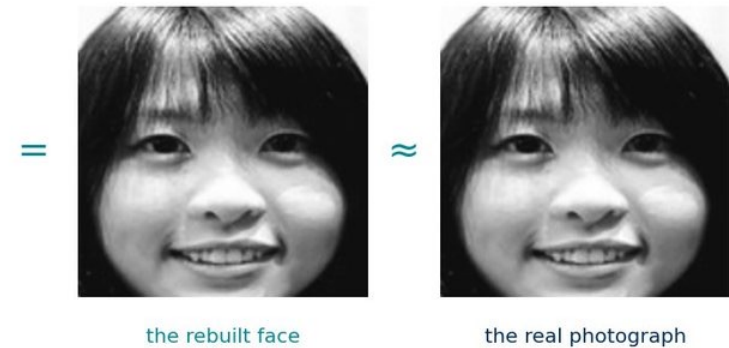
Twenty-seven eigenfaces

Twenty-seven. The last three are worth 0.2% of the error between them.



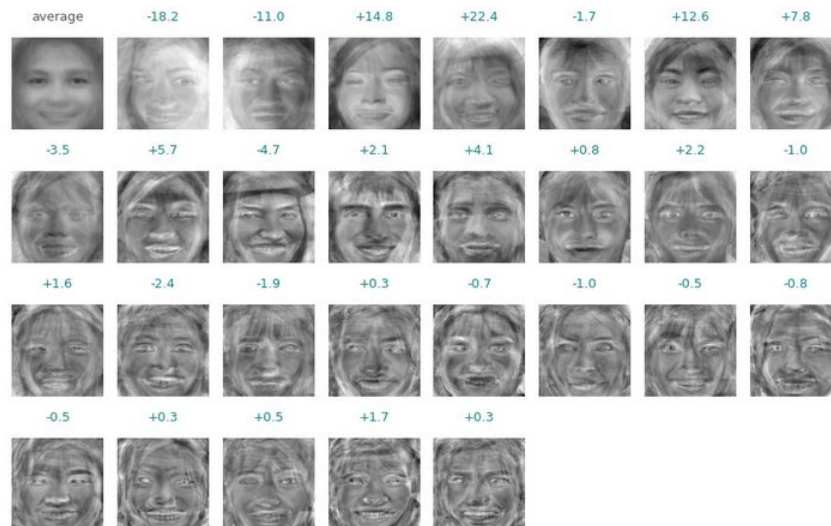
E1 to E27 in reading order — the teal number over each one is how much of it this face needs

average pixel still 0.2% off

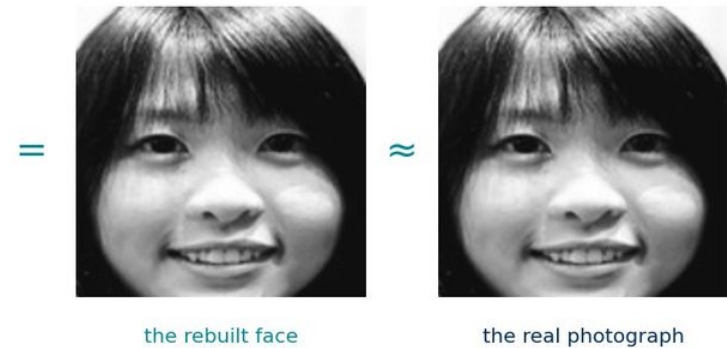


Twenty-eight eigenfaces

Twenty-eight terms. The picture has stopped changing; only the error percentage is still moving.



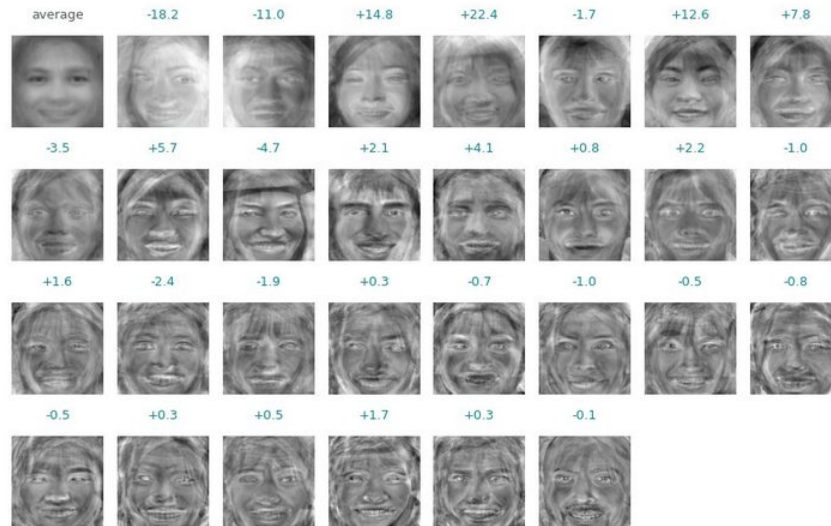
E1 to E28 in reading order — the teal number over each one is how much of it this face needs



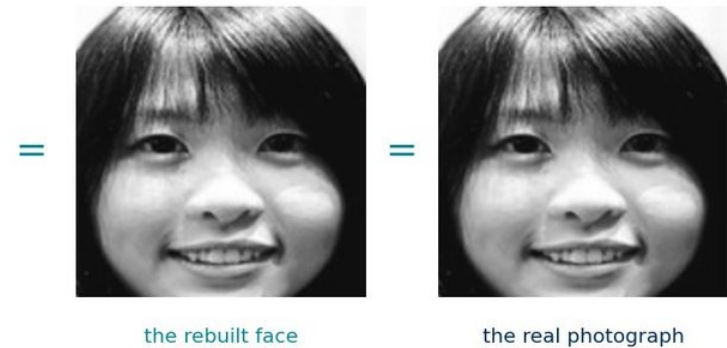
average pixel still 0.1% off

Twenty-nine eigenfaces

All twenty-nine. The approximation has become the photograph exactly — and the last twenty-one eigenfaces, more than two thirds of the set, did only 7% of the work.



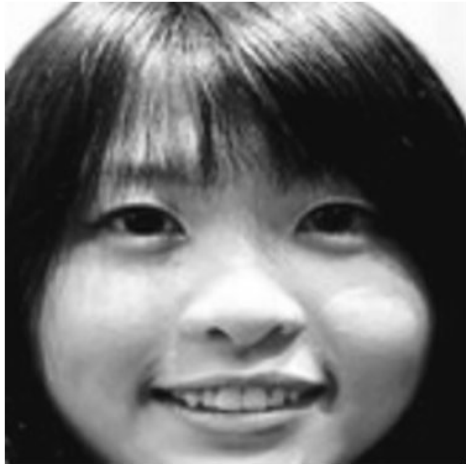
E1 to E29 in reading order — the teal number over each one is how much of it this face needs



the rebuild is now exact

100% of her captured

The whole sequence, in one table

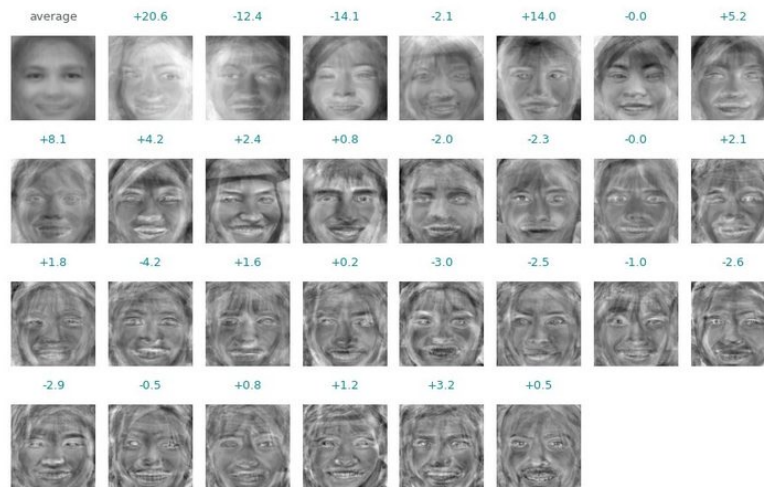


after k	average pixel error	how much of her is captured
1	22.5%	22%
2	21.7%	30%
4	11.7%	78%
6	7.9%	88%
8	6.1%	93%
15	2.7%	99%
20	1.4%	100%
29	0% — exact	100%

Read the last column rather than the first: doubling the eigenfaces from eight to fifteen buys six percentage points of match, and the fourteen after that buy one.

The same recipe, a different person

A different face, the identical twenty-nine eigenfaces. Watch the numbers above them rather than the pictures: his E1 amount is +20.6 where hers was -18.2 — opposite signs, because he is lighter than the average face and she is darker.



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the rebuilt face

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the real photograph

E1 to E29 in reading order — the teal number over each one is how much of it this face needs

nothing on the left-hand side is new — only the numbers above the eigenfaces

One eigenface

His amount of E1 is +20.6, where hers was -18.2. Opposite signs: he sits on the bright side of the average, she on the dark.

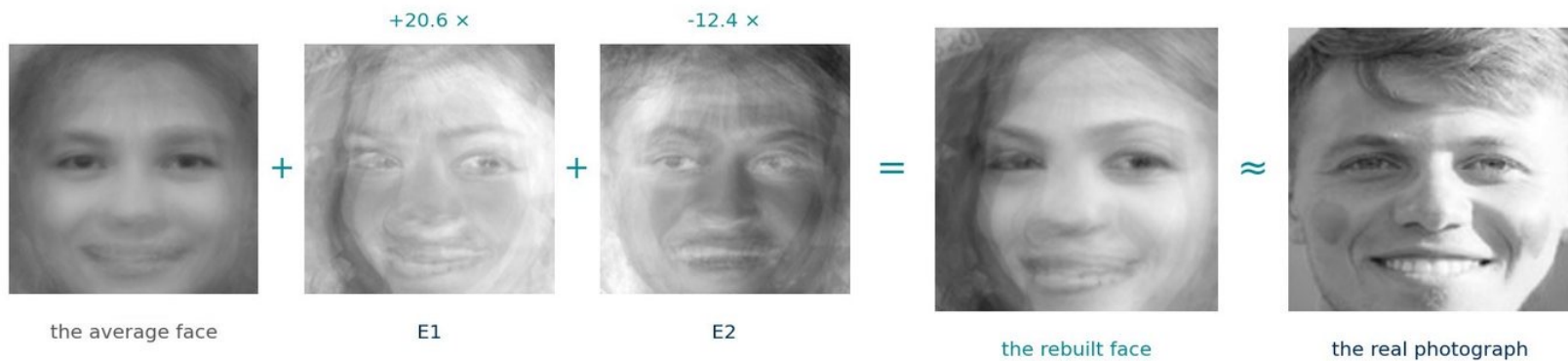


average pixel still 18% off

36% captured

Two eigenfaces

Two terms. The average pixel is 16.5% off, and 49% of him is captured.

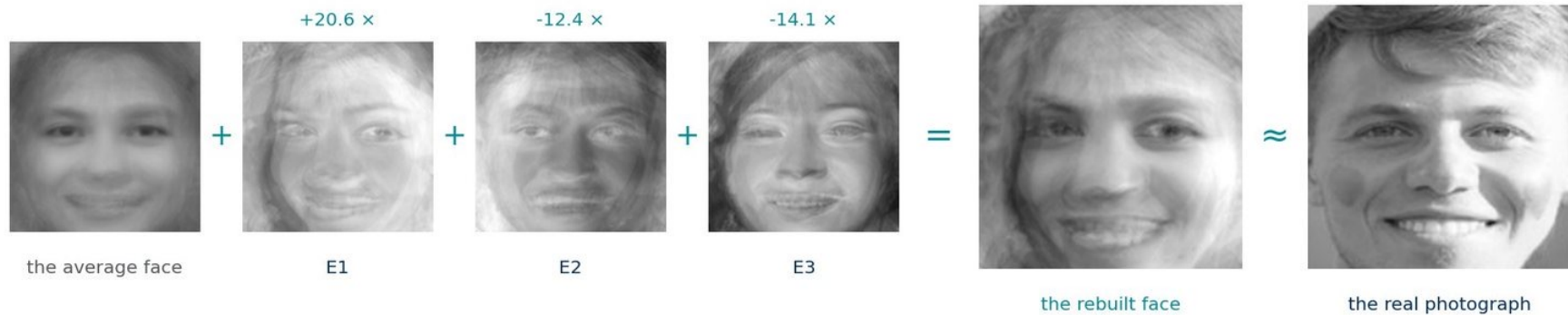


average pixel still 16.5% off

49% captured

Three eigenfaces

Three terms. The average pixel is 12.8% off, and 66% of him is captured.



average pixel still 12.8% off

66% captured

Four eigenfaces

Four terms. 12.8% off, 66% of him captured — compare that with her 78% at the same point.

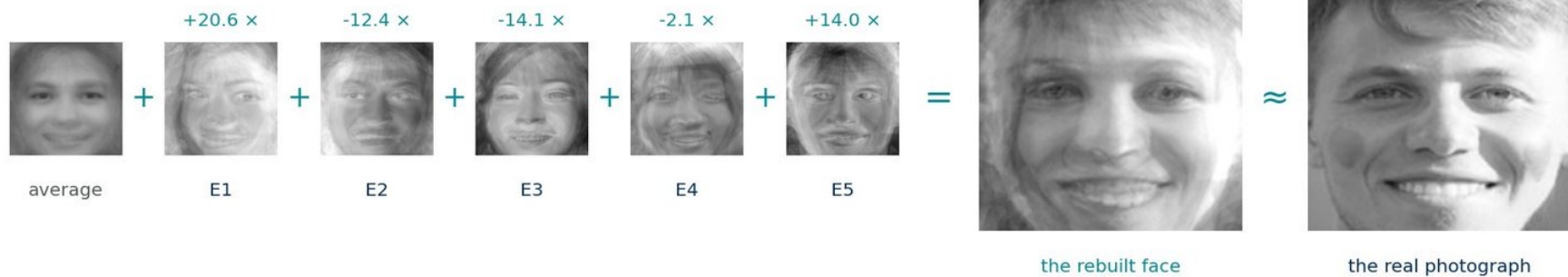


average pixel still 12.8% off

66% captured

Five eigenfaces

Five terms. The average pixel is 8.7% off, and 83% of him is captured.



average pixel still 8.7% off

83% captured

Six eigenfaces

Six terms. The average pixel is 8.7% off, and 83% of him is captured.



E1 to E6 in reading order — the teal number over each one is how much of it this face needs



average pixel still 8.7% off

83% captured

Seven eigenfaces

Seven terms. The average pixel is 8.1% off, and 85% of him is captured.



E1 to E7 in reading order — the teal number over each one is how much of it this face needs

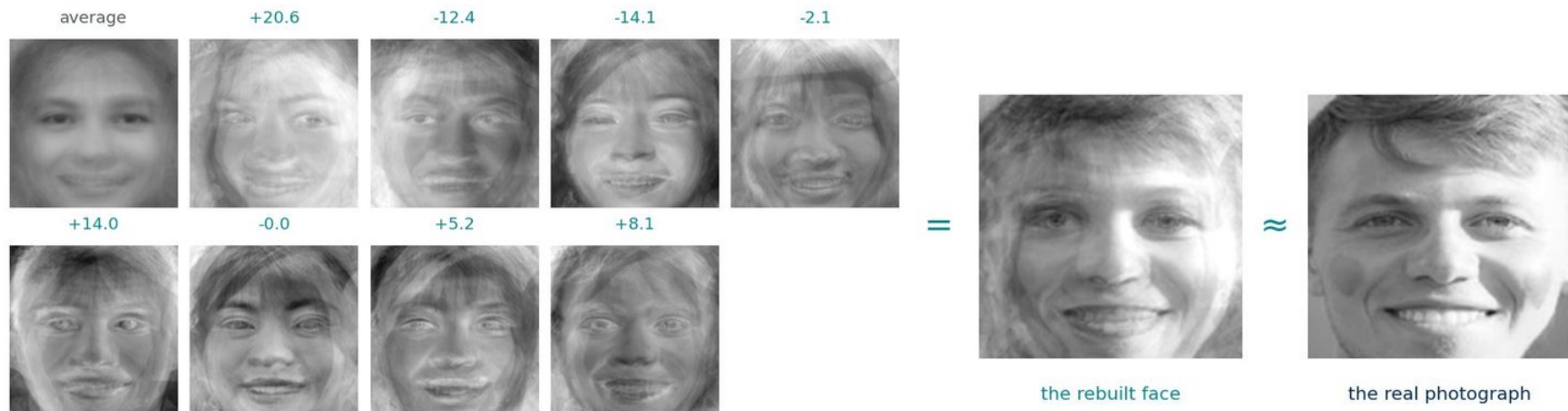


average pixel still 8.1% off

85% captured

Eight eigenfaces

Eight eigenfaces carry 91% of him. A different person, a different set of amounts, the same collapse.

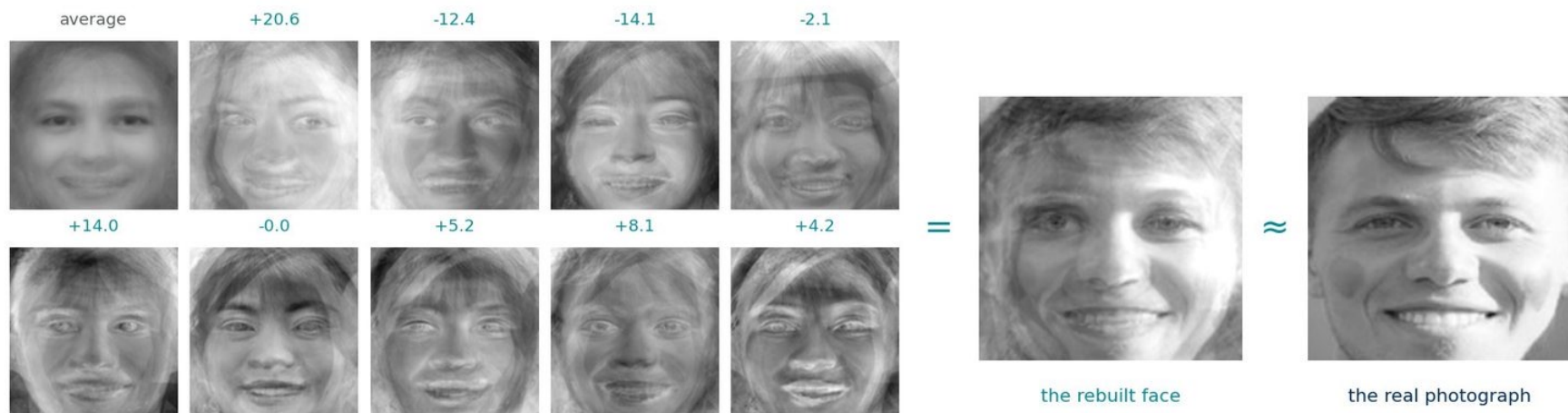


E1 to E8 in reading order — the teal number over each one is how much of it this face needs

average pixel still 6.1% off

Nine eigenfaces

Nine terms. The average pixel is 5.7% off, and 93% of him is captured.



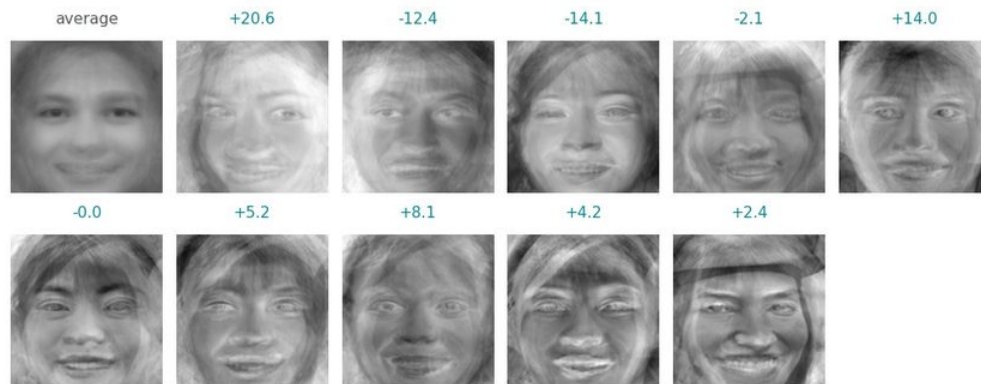
E1 to E9 in reading order — the teal number over each one is how much of it this face needs

average pixel still 5.7% off

93% captured

Ten eigenfaces

Ten terms. The average pixel is 5.5% off, and 93% of him is captured.



E1 to E10 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

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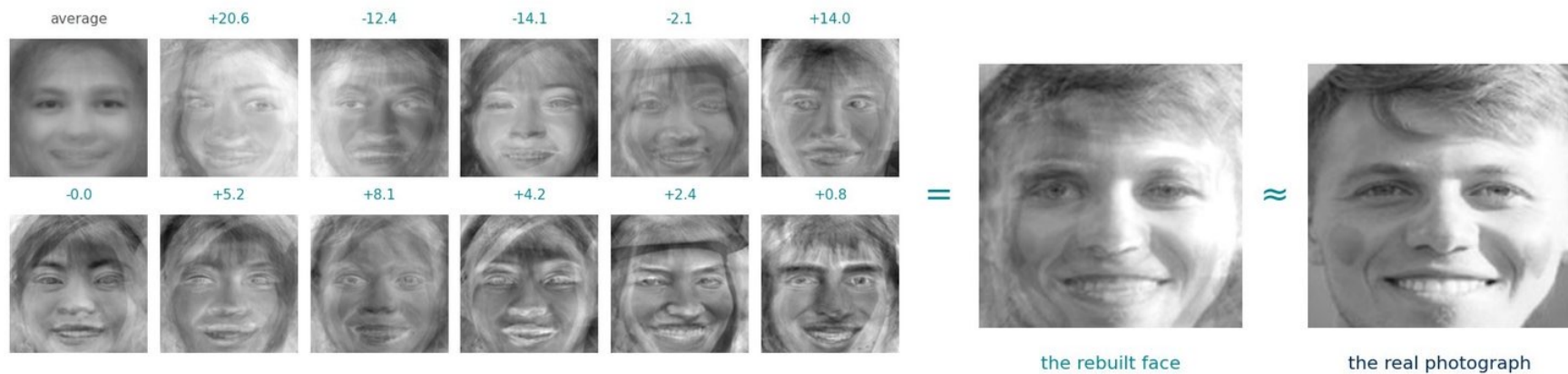
the real photograph

average pixel still 5.5% off

93% captured

Eleven eigenfaces

Eleven terms. The average pixel is 5.5% off, and 93% of him is captured.



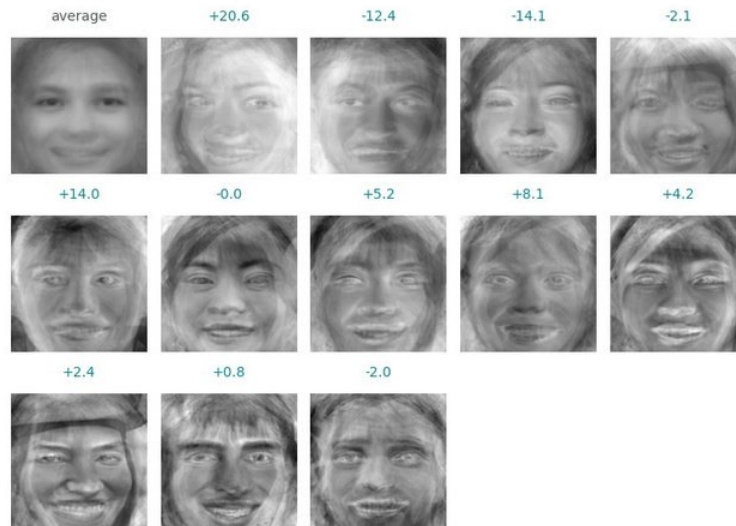
E1 to E11 in reading order — the teal number over each one is how much of it this face needs

average pixel still 5.5% off

93% captured

Twelve eigenfaces

Twelve terms. The average pixel is 5.4% off, and 93% of him is captured.



E1 to E12 in reading order — the teal number over each one is how much of it this face needs

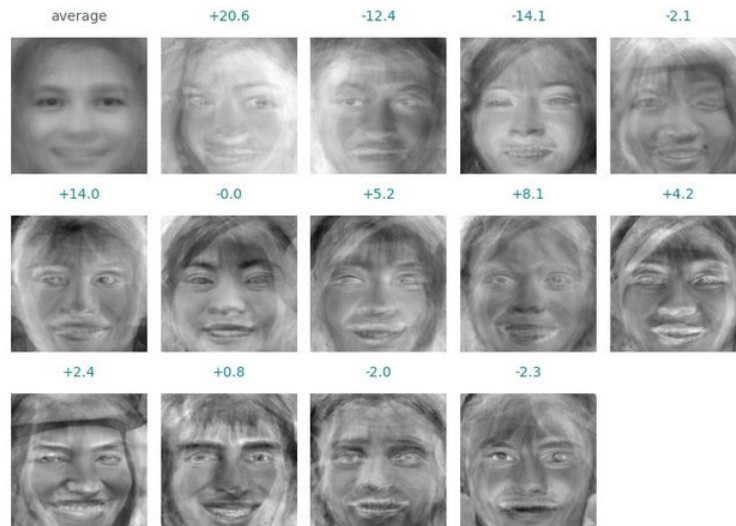


average pixel still 5.4% off

93% captured

Thirteen eigenfaces

Thirteen terms. The average pixel is 5.2% off, and 94% of him is captured.



E1 to E13 in reading order — the teal number over each one is how much of it this face needs

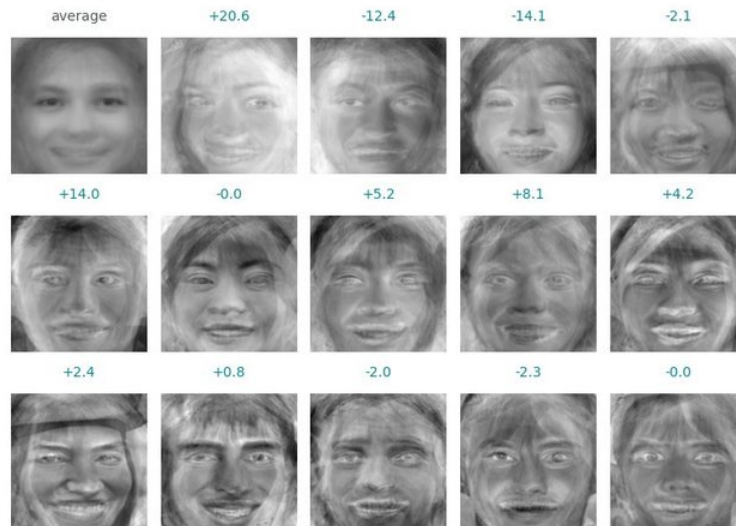


average pixel still 5.2% off

94% captured

Fourteen eigenfaces

Fourteen terms. The average pixel is 5.2% off, and 94% of him is captured.



E1 to E14 in reading order — the teal number over each one is how much of it this face needs

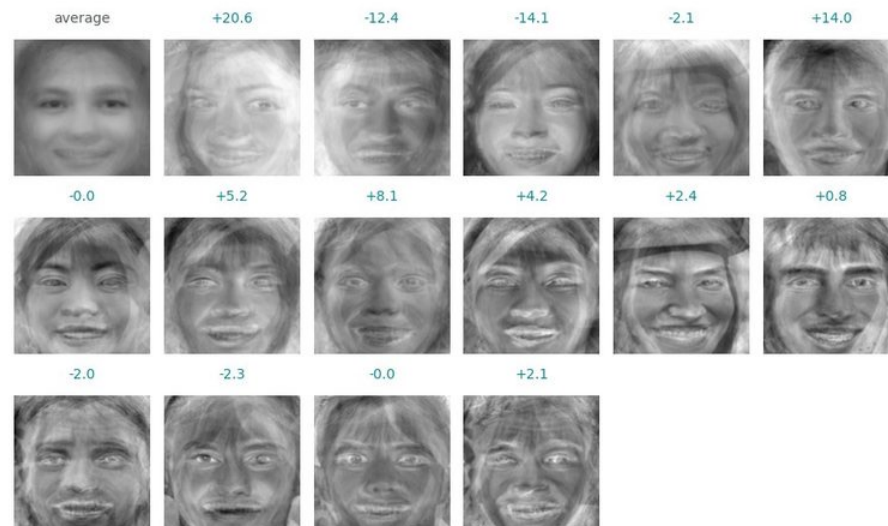


average pixel still 5.2% off

94% captured

Fifteen eigenfaces

Fifteen. 94% captured and 5.1% off. The rest is texture.



E1 to E15 in reading order — the teal number over each one is how much of it this face needs



average pixel still 5.1% off

Sixteen eigenfaces

Sixteen terms. The average pixel is 4.9% off, and 95% of him is captured.



E1 to E16 in reading order — the teal number over each one is how much of it this face needs

average pixel still 4.9% off

95% captured

Seventeen eigenfaces

Seventeen terms. The average pixel is 4.2% off, and 96% of him is captured.



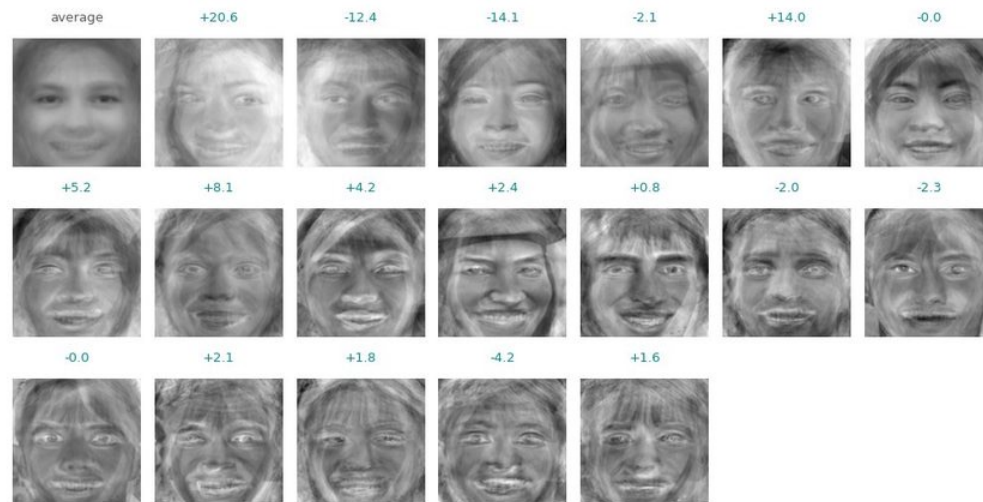
E1 to E17 in reading order — the teal number over each one is how much of it this face needs

average pixel still 4.2% off

96% captured

Eighteen eigenfaces

Eighteen terms. The average pixel is 4% off, and 96% of him is captured.



=



the rebuilt face

≈



the real photograph

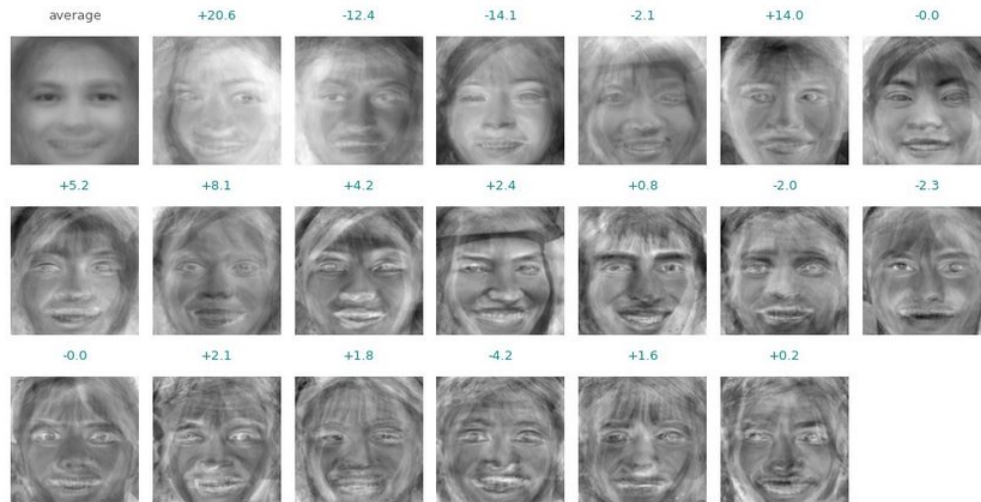
E1 to E18 in reading order — the teal number over each one is how much of it this face needs

average pixel still 4% off

96% captured

Nineteen eigenfaces

Nineteen terms. The average pixel is 4% off, and 96% of him is captured.



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the rebuilt face

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the real photograph

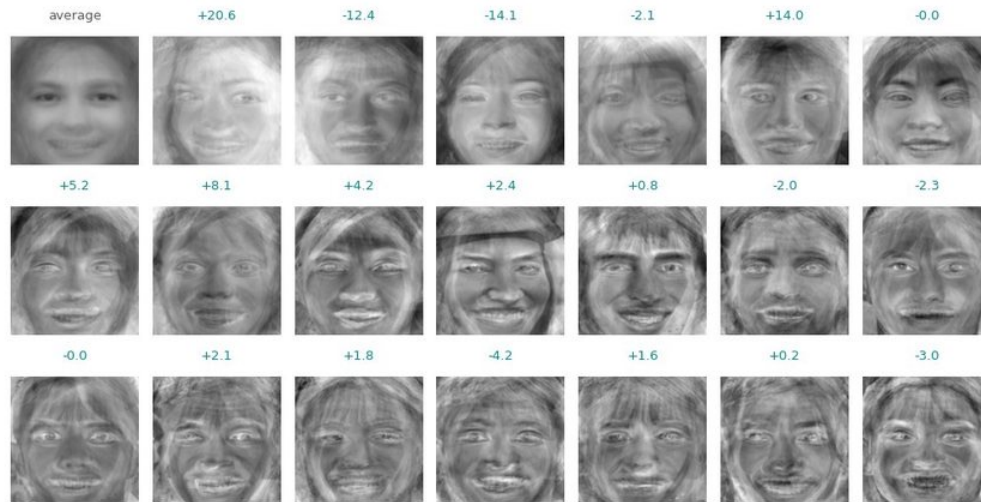
E1 to E19 in reading order — the teal number over each one is how much of it this face needs

average pixel still 4% off

96% captured

Twenty eigenfaces

Twenty. 3.7% off. Nothing you can see is still changing.



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the rebuilt face

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the real photograph

E1 to E20 in reading order — the teal number over each one is how much of it this face needs

average pixel still 3.7% off

97% captured

Twenty-one eigenfaces

Twenty-one terms. The average pixel is 3.3% off, and 98% of him is captured.



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the rebuilt face

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the real photograph

E1 to E21 in reading order — the teal number over each one is how much of it this face needs

average pixel still 3.3% off

98% captured

Twenty-two eigenfaces

Twenty-two terms. The average pixel is 3.2% off, and 98% of him is captured.



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the rebuilt face

≈



the real photograph

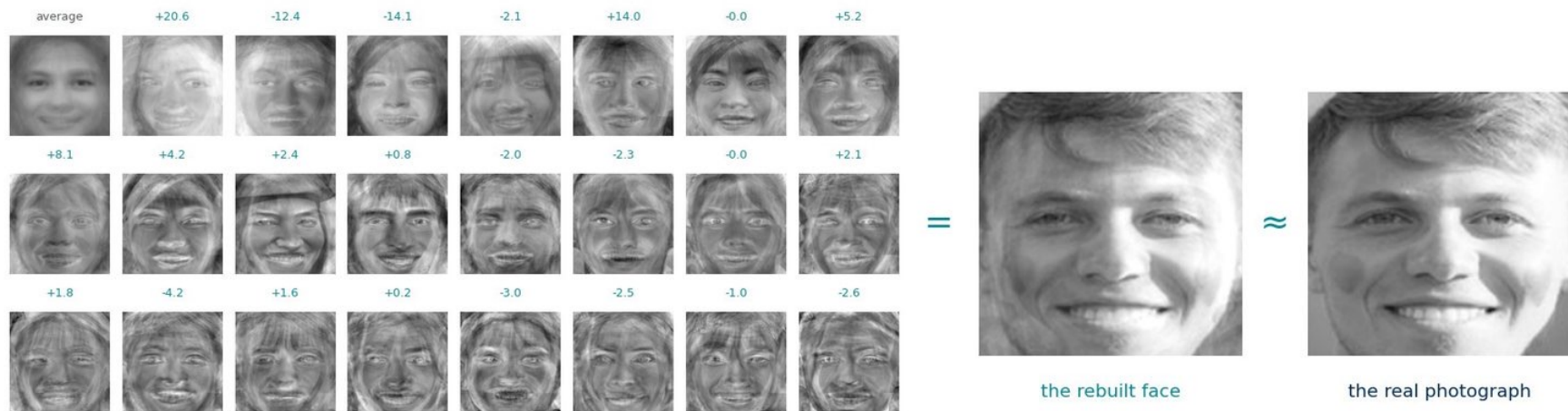
E1 to E22 in reading order — the teal number over each one is how much of it this face needs

average pixel still 3.2% off

98% captured

Twenty-three eigenfaces

Twenty-three terms. The average pixel is 2.7% off, and 98% of him is captured.



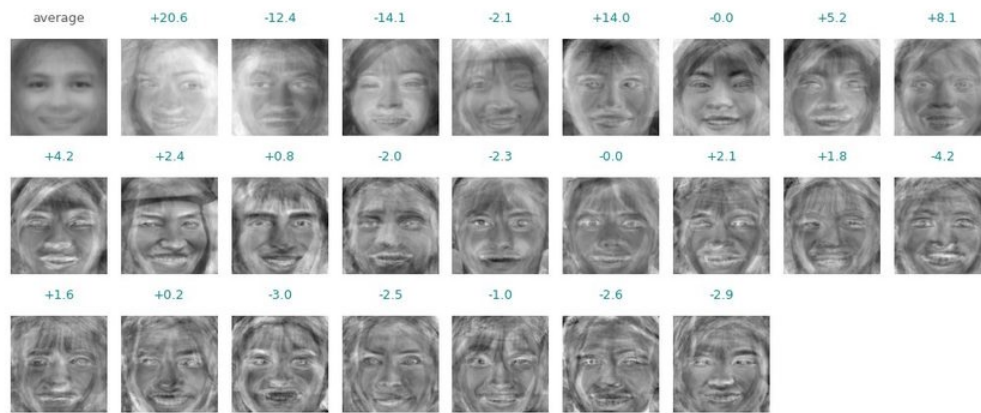
E1 to E23 in reading order — the teal number over each one is how much of it this face needs

average pixel still 2.7% off

98% captured

Twenty-four eigenfaces

Twenty-four terms. The average pixel is 2.2% off, and 99% of him is captured.



E1 to E24 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

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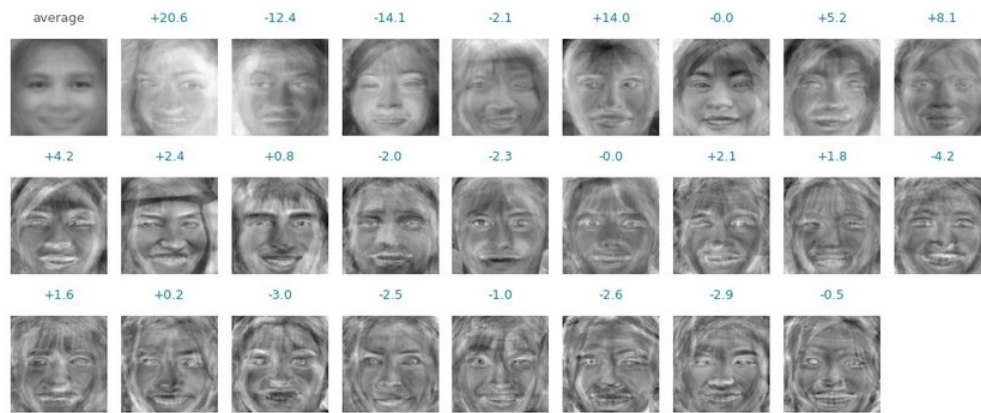
the real photograph

average pixel still 2.2% off

99% captured

Twenty-five eigenfaces

Twenty-five terms. The average pixel is 2.2% off, and 99% of him is captured.



E1 to E25 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

≈



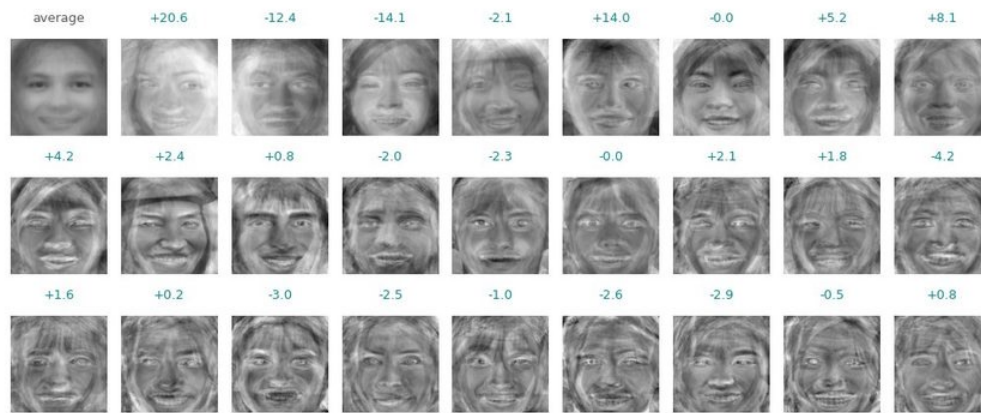
the real photograph

average pixel still 2.2% off

99% captured

Twenty-six eigenfaces

Twenty-six terms. The average pixel is 2.1% off, and 99% of him is captured.



E1 to E26 in reading order — the teal number over each one is how much of it this face needs

=



the rebuilt face

≈



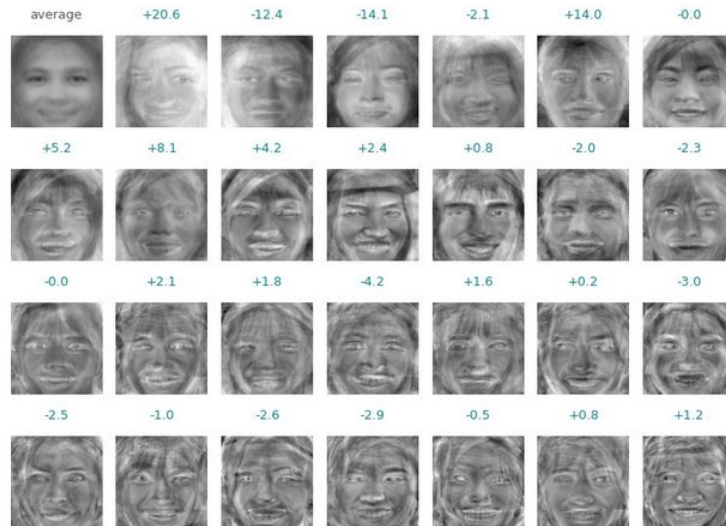
the real photograph

average pixel still 2.1% off

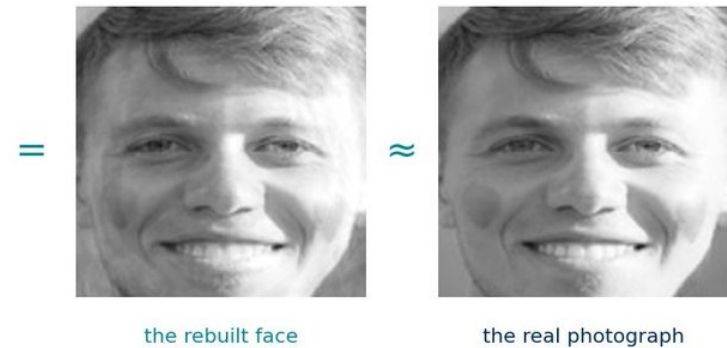
99% captured

Twenty-seven eigenfaces

Twenty-seven terms. The average pixel is 2% off, and 99% of him is captured.



E1 to E27 in reading order — the teal number over each one is how much of it this face needs

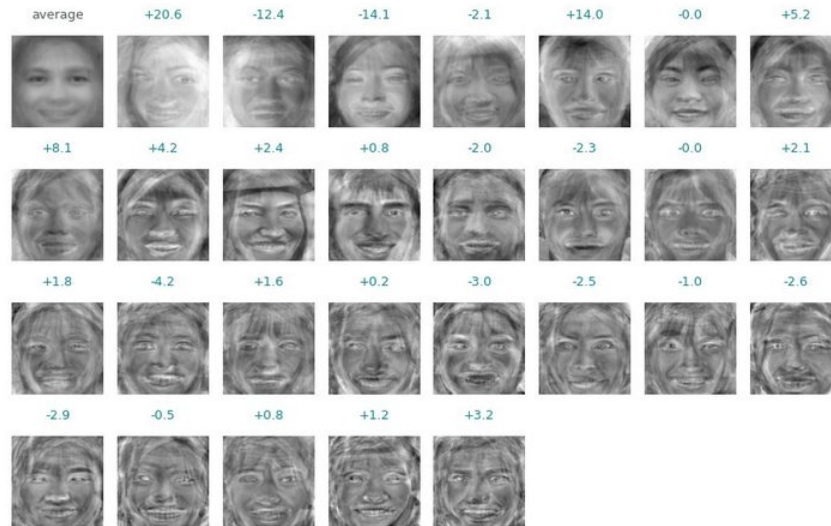


average pixel still 2% off

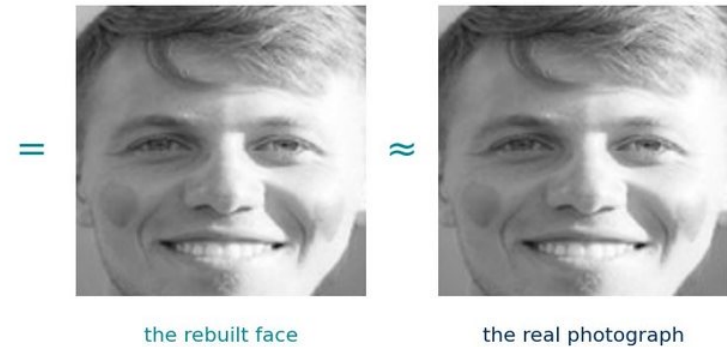
99% captured

Twenty-eight eigenfaces

Twenty-eight terms. The average pixel is 0.3% off, and 100% of him is captured.



E1 to E28 in reading order — the teal number over each one is how much of it this face needs

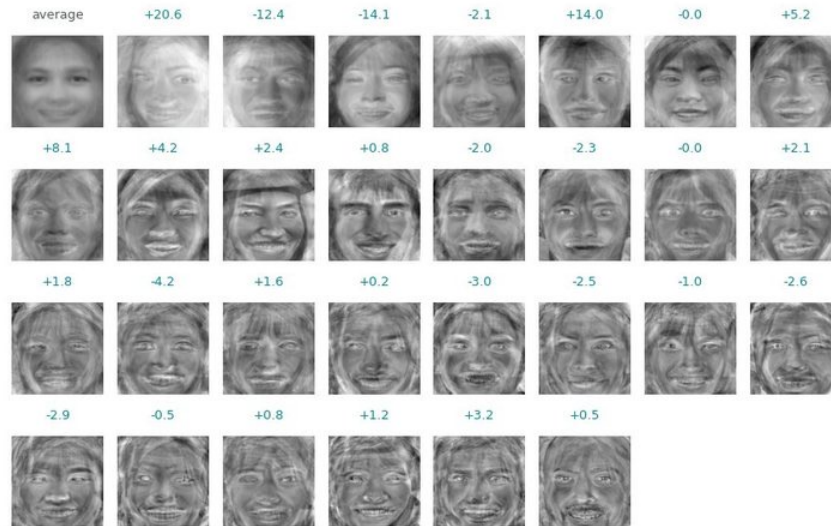


average pixel still 0.3% off

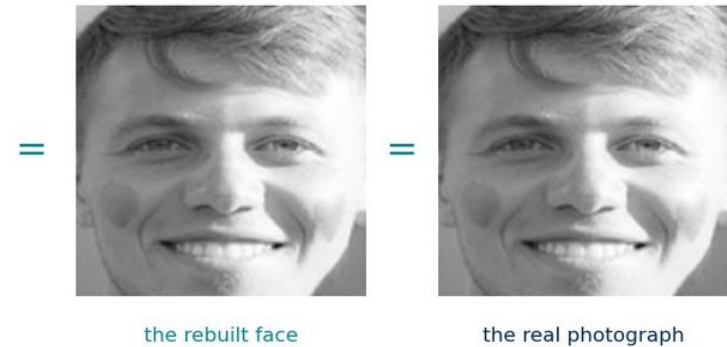
100% captured

Twenty-nine eigenfaces

All twenty-nine, exact. He needed different numbers from her, but the same short list of them.



E1 to E29 in reading order — the teal number over each one is how much of it this face needs



the rebuild is now exact

100% captured

The whole sequence, in one table

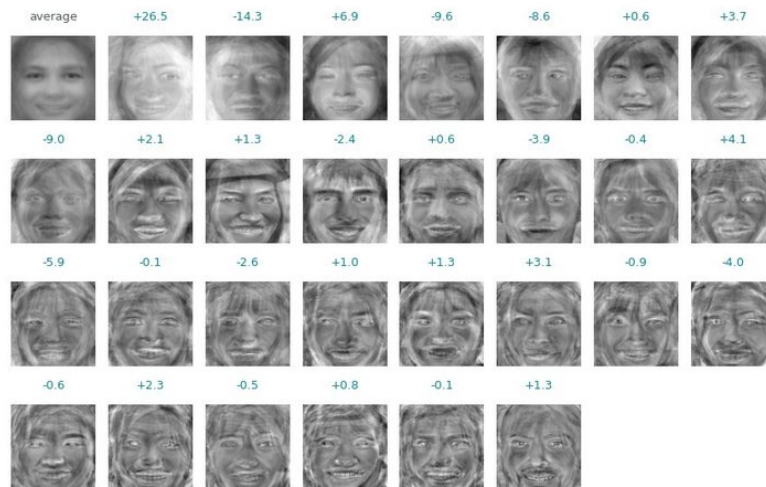


after k	average pixel error	how much of him is captured
1	18%	36%
2	16.5%	49%
4	12.8%	66%
6	8.7%	83%
8	6.1%	91%
15	5.1%	94%
20	3.7%	97%
29	0% — exact	100%

The same shape of curve on a different person — steep to about eight, then flat. Note he starts further ahead: E1 alone already captures 36% of him against 22% of her.

The same recipe, a different person

One more. By now the format should be doing the explaining: an average, twenty-nine fixed patterns, and one number each. This face has the largest first amount of the three.



E1 to E29 in reading order — the teal number over each one is how much of it this face needs

nothing on the left-hand side is new — only the numbers above the eigenfaces

One eigenface

Her E1 amount is +26.5 — the largest of the three examples. She is the brightest face against the average.



average pixel still 14.6% off

53% captured

Two eigenfaces

Two terms. The average pixel is 12.9% off, and 68% of her is captured.

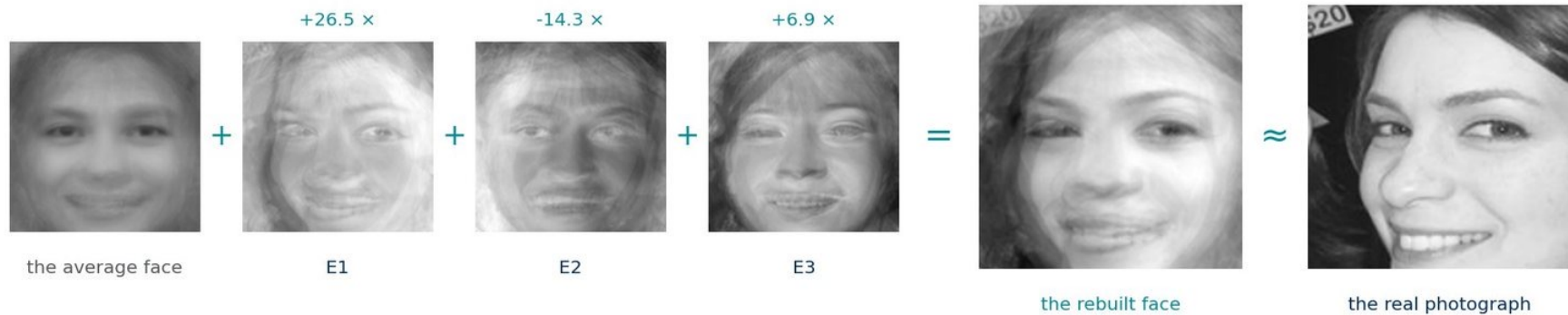


average pixel still 12.9% off

68% captured

Three eigenfaces

Three terms. The average pixel is 11.4% off, and 71% of her is captured.



average pixel still 11.4% off

71% captured

Four eigenfaces

Four terms, 78% of her. The head tilt is what the early eigenfaces struggle with here.

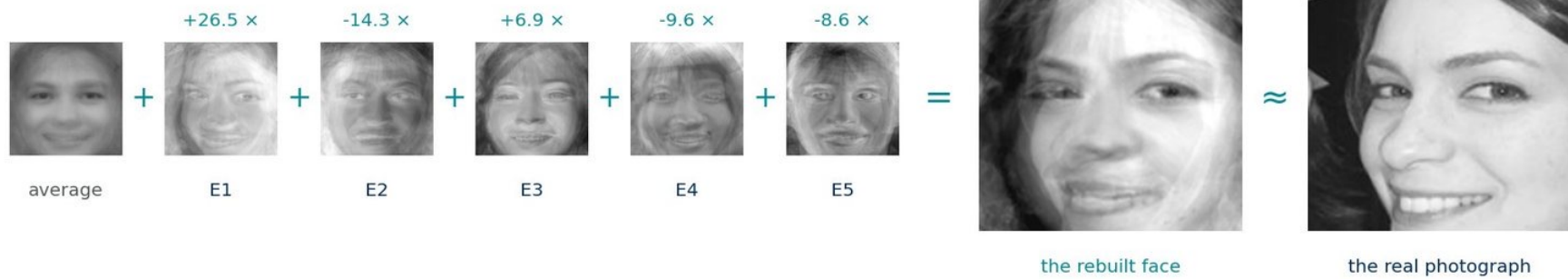


average pixel still 10.3% off

78% captured

Five eigenfaces

Five terms. The average pixel is 9% off, and 84% of her is captured.



average pixel still 9% off

84% captured

Six eigenfaces

Six terms. The average pixel is 8.9% off, and 84% of her is captured.



E1 to E6 in reading order — the teal number over each one is how much of it this face needs



average pixel still 8.9% off

84% captured

Seven eigenfaces

Seven terms. The average pixel is 8.5% off, and 85% of her is captured.



E1 to E7 in reading order — the teal number over each one is how much of it this face needs



average pixel still 8.5% off

85% captured

Eight eigenfaces

Eight eigenfaces, 91% captured, 6.5% off.



=



the rebuilt face

≈



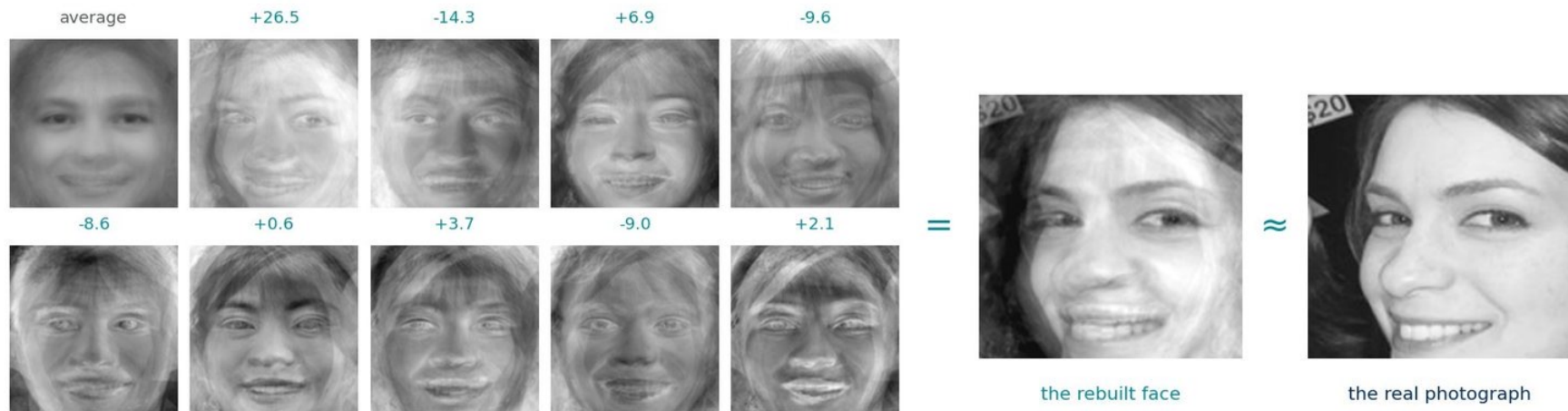
the real photograph

E1 to E8 in reading order — the teal number over each one is how much of it this face needs

average pixel still 6.5% off

Nine eigenfaces

Nine terms. The average pixel is 6.4% off, and 91% of her is captured.



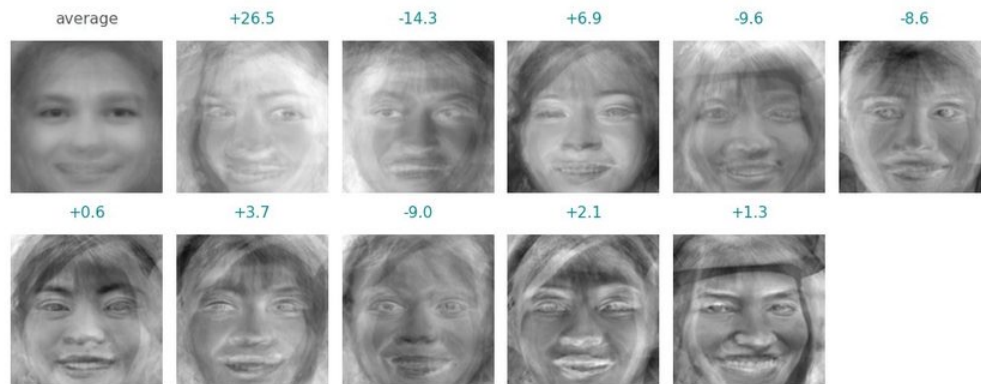
E1 to E9 in reading order — the teal number over each one is how much of it this face needs

average pixel still 6.4% off

91% captured

Ten eigenfaces

Ten terms. The average pixel is 6.3% off, and 91% of her is captured.



E1 to E10 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

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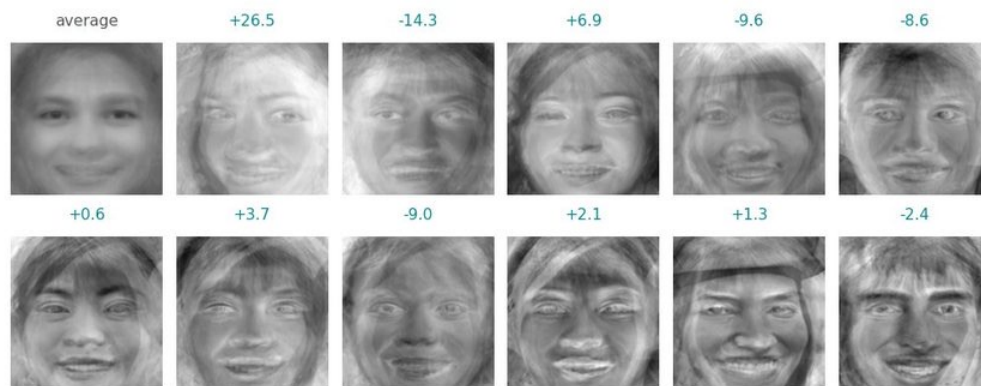
the real photograph

average pixel still 6.3% off

91% captured

Eleven eigenfaces

Eleven terms. The average pixel is 6.1% off, and 92% of her is captured.



E1 to E11 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

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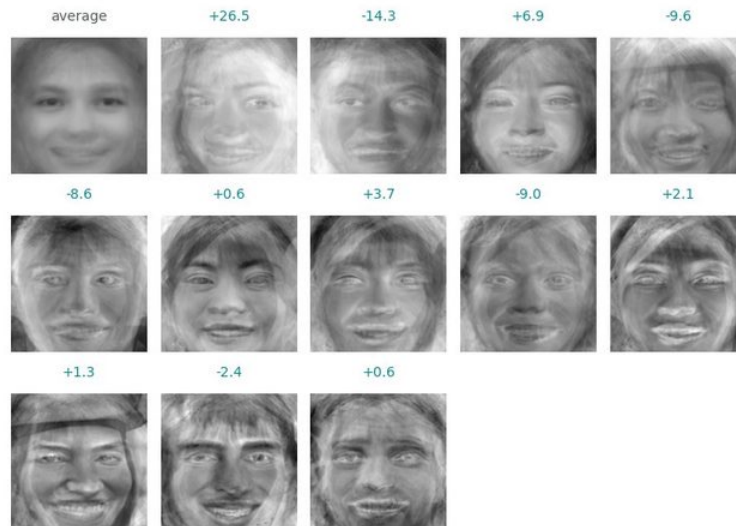
the real photograph

average pixel still 6.1% off

92% captured

Twelve eigenfaces

Twelve terms. The average pixel is 6.1% off, and 92% of her is captured.



E1 to E12 in reading order — the teal number over each one is how much of it this face needs

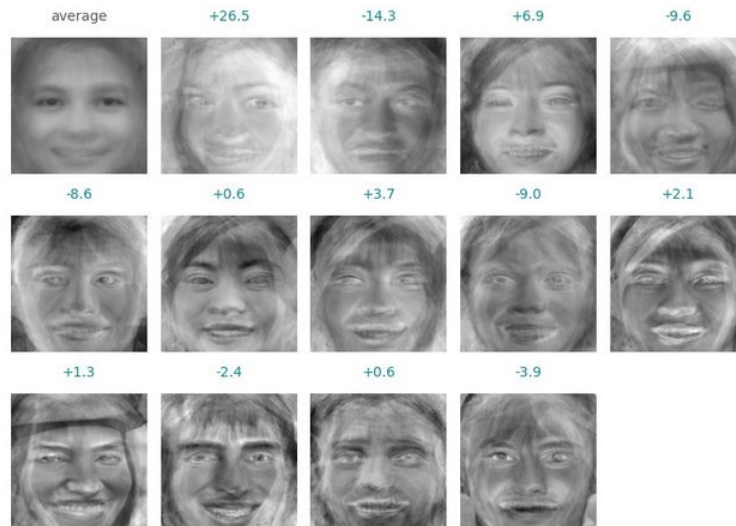
average pixel still 6.1% off



92% captured

Thirteen eigenfaces

Thirteen terms. The average pixel is 5.5% off, and 93% of her is captured.



E1 to E13 in reading order — the teal number over each one is how much of it this face needs



average pixel still 5.5% off

93% captured

Fourteen eigenfaces

Fourteen terms. The average pixel is 5.5% off, and 93% of her is captured.



E1 to E14 in reading order — the teal number over each one is how much of it this face needs

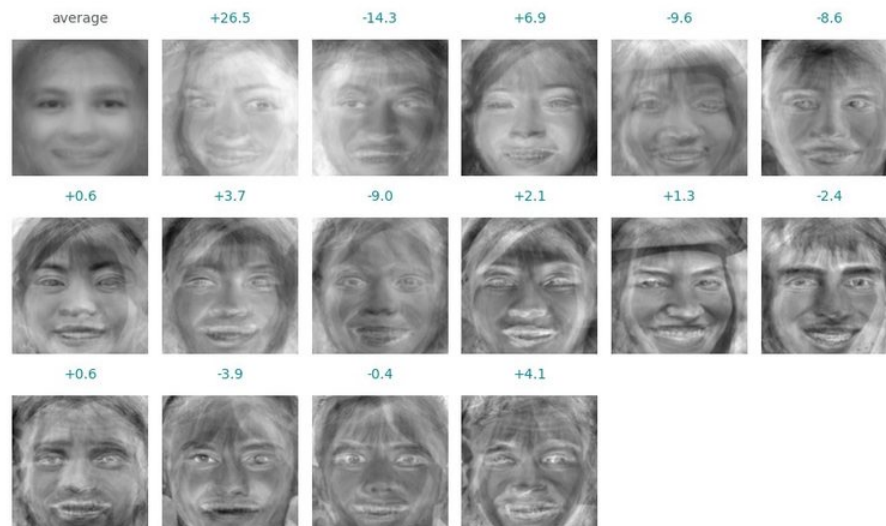


average pixel still 5.5% off

93% captured

Fifteen eigenfaces

Fifteen. 94% and 5.2% off.



E1 to E15 in reading order — the teal number over each one is how much of it this face needs

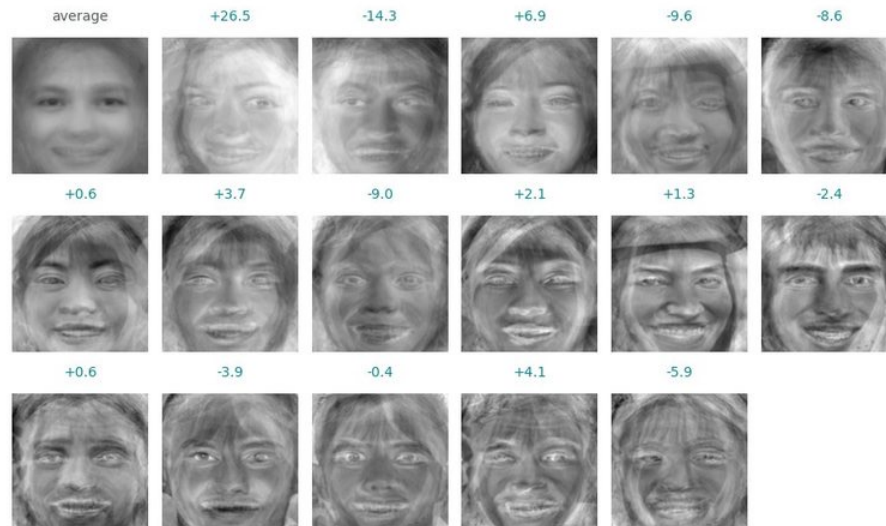


average pixel still 5.2% off

94% captured

Sixteen eigenfaces

Sixteen terms. The average pixel is 4% off, and 97% of her is captured.



E1 to E16 in reading order — the teal number over each one is how much of it this face needs

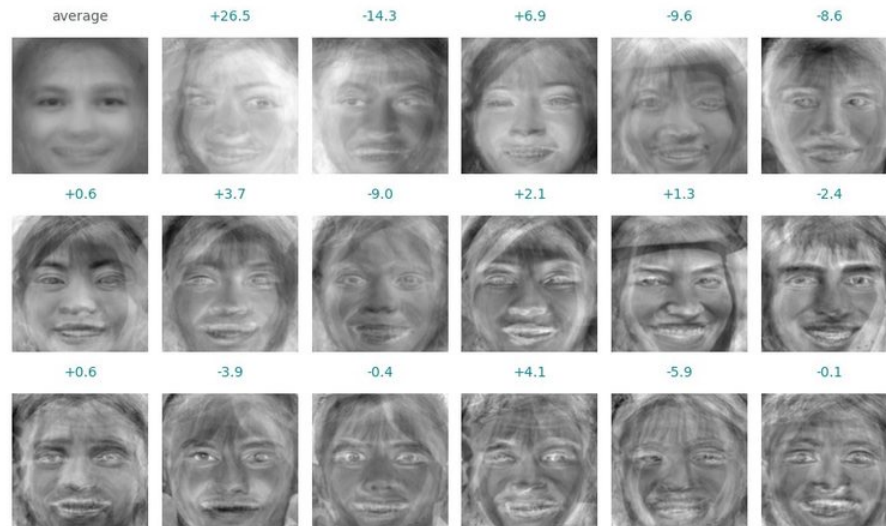
average pixel still 4% off



97% captured

Seventeen eigenfaces

Seventeen terms. The average pixel is 4% off, and 97% of her is captured.



E1 to E17 in reading order — the teal number over each one is how much of it this face needs

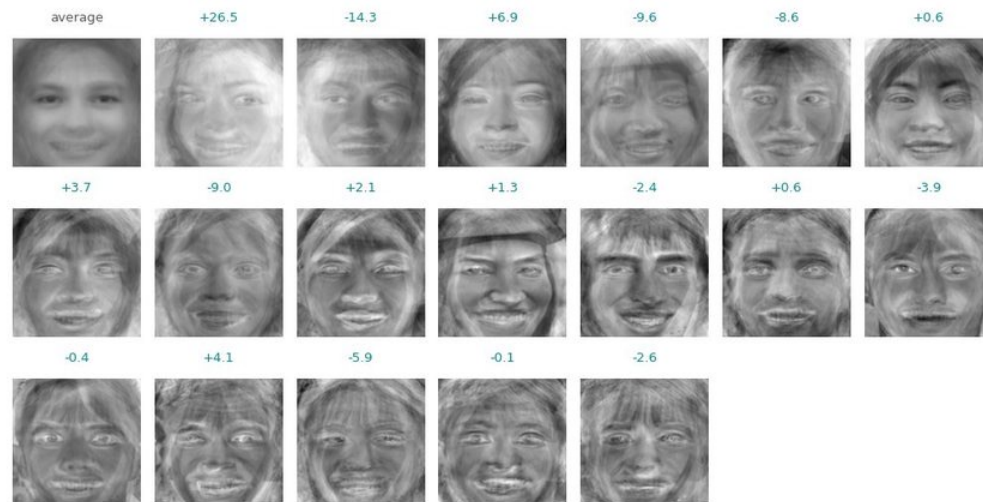


average pixel still 4% off

97% captured

Eighteen eigenfaces

Eighteen terms. The average pixel is 3.6% off, and 97% of her is captured.



=



the rebuilt face

≈



the real photograph

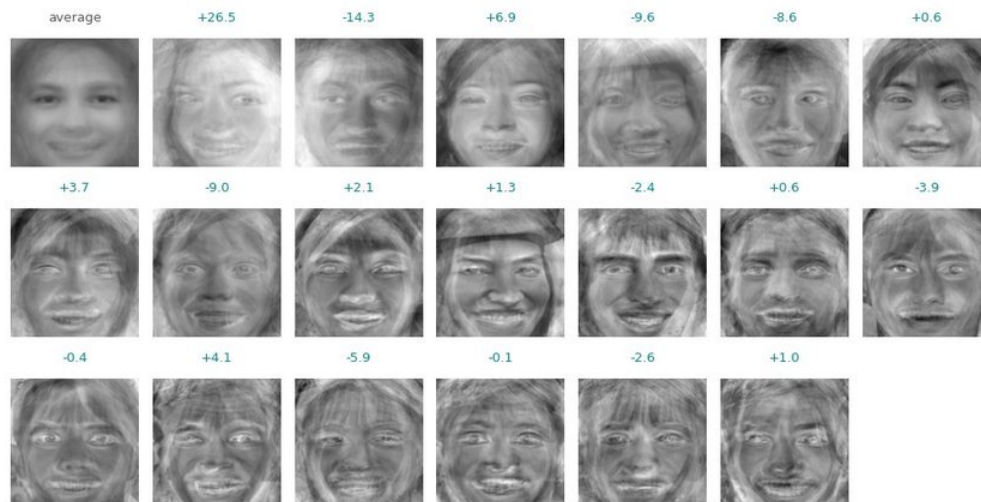
E1 to E18 in reading order — the teal number over each one is how much of it this face needs

average pixel still 3.6% off

97% captured

Nineteen eigenfaces

Nineteen terms. The average pixel is 3.6% off, and 97% of her is captured.



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the rebuilt face

≈



the real photograph

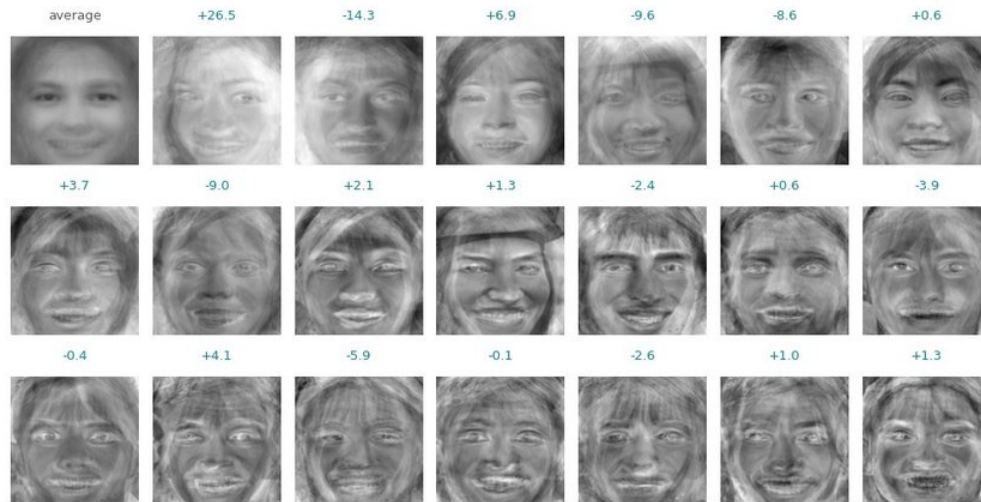
E1 to E19 in reading order — the teal number over each one is how much of it this face needs

average pixel still 3.6% off

97% captured

Twenty eigenfaces

Twenty. 3.5% off — and the last nine will take three quarters of an hour of your class to click through for almost no visible gain.



=



the rebuilt face

≈



the real photograph

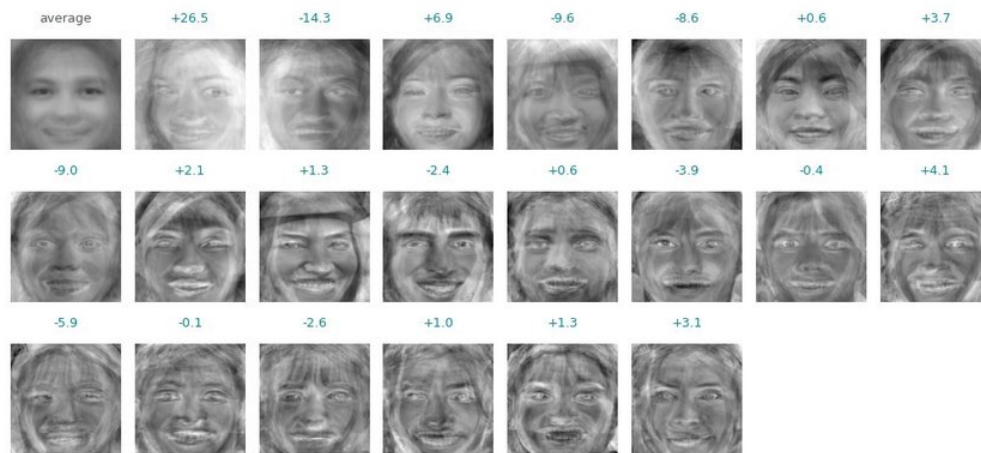
E1 to E20 in reading order — the teal number over each one is how much of it this face needs

average pixel still 3.5% off

97% captured

Twenty-one eigenfaces

Twenty-one terms. The average pixel is 3% off, and 98% of her is captured.



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the rebuilt face

≈



the real photograph

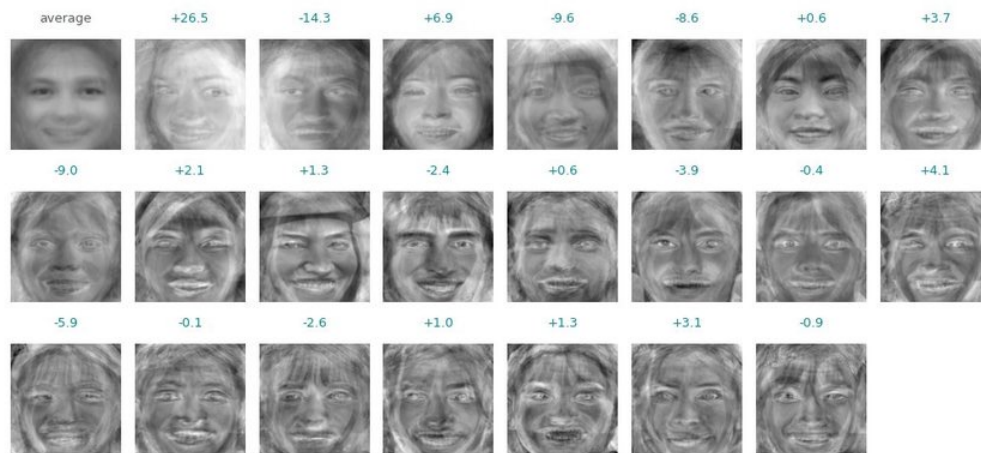
E1 to E21 in reading order — the teal number over each one is how much of it this face needs

average pixel still 3% off

98% captured

Twenty-two eigenfaces

Twenty-two terms. The average pixel is 2.9% off, and 98% of her is captured.



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the rebuilt face

≈



the real photograph

E1 to E22 in reading order — the teal number over each one is how much of it this face needs

average pixel still 2.9% off

98% captured

Twenty-three eigenfaces

Twenty-three terms. The average pixel is 1.7% off, and 99% of her is captured.



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the rebuilt face

≈



the real photograph

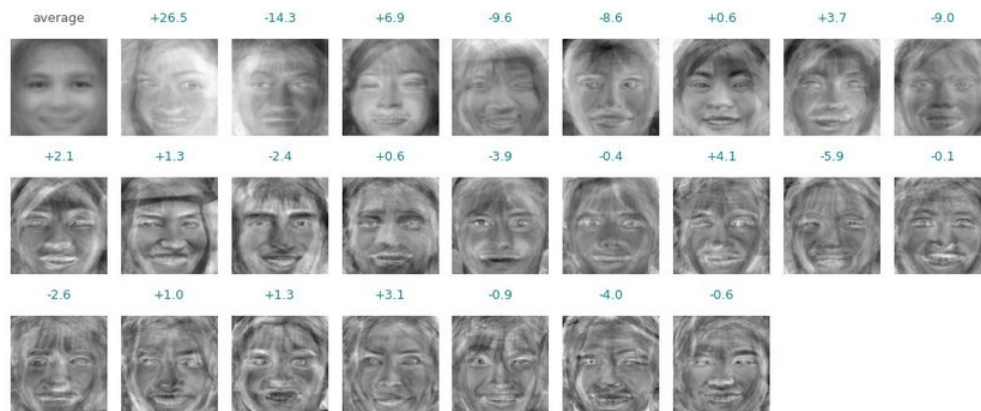
E1 to E23 in reading order — the teal number over each one is how much of it this face needs

average pixel still 1.7% off

99% captured

Twenty-four eigenfaces

Twenty-four terms. The average pixel is 1.7% off, and 99% of her is captured.



E1 to E24 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

≈



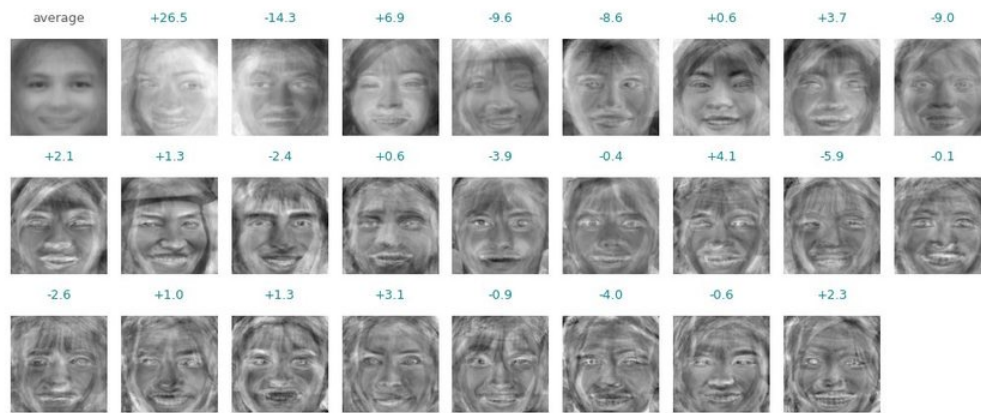
the real photograph

average pixel still 1.7% off

99% captured

Twenty-five eigenfaces

Twenty-five terms. The average pixel is 0.9% off, and 100% of her is captured.



E1 to E25 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

≈



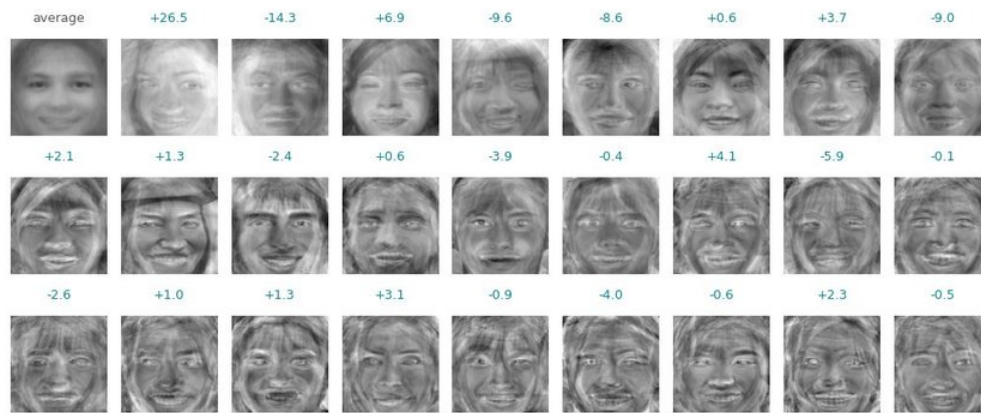
the real photograph

average pixel still 0.9% off

100% captured

Twenty-six eigenfaces

Twenty-six terms. The average pixel is 0.9% off, and 100% of her is captured.



E1 to E26 in reading order — the teal number over each one is how much of it this face needs

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the rebuilt face

≈



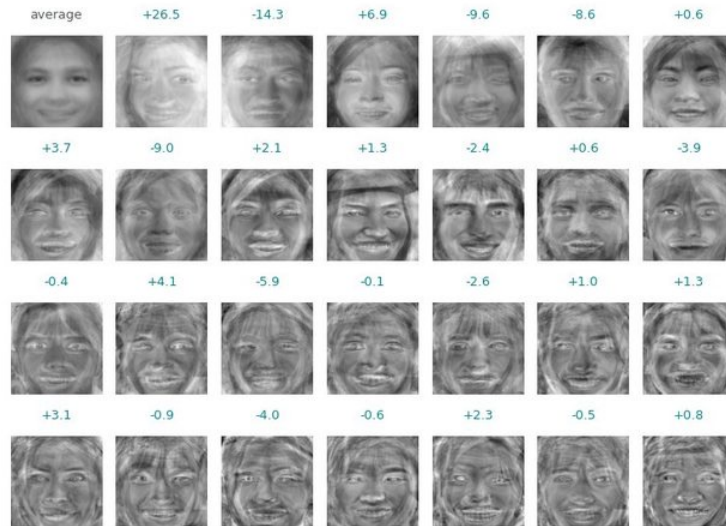
the real photograph

average pixel still 0.9% off

100% captured

Twenty-seven eigenfaces

Twenty-seven terms. The average pixel is 0.8% off, and 100% of her is captured.



E1 to E27 in reading order — the teal number over each one is how much of it this face needs

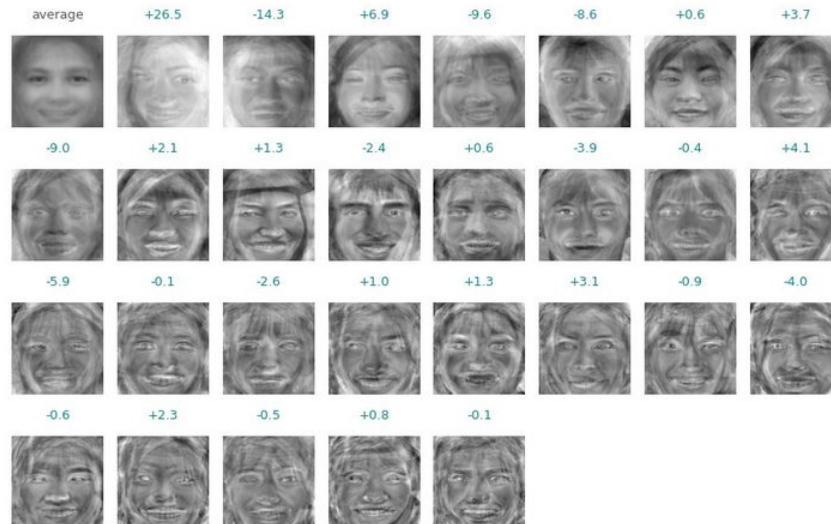
average pixel still 0.8% off



100% captured

Twenty-eight eigenfaces

Twenty-eight terms. The average pixel is 0.8% off, and 100% of her is captured.



E1 to E28 in reading order — the teal number over each one is how much of it this face needs

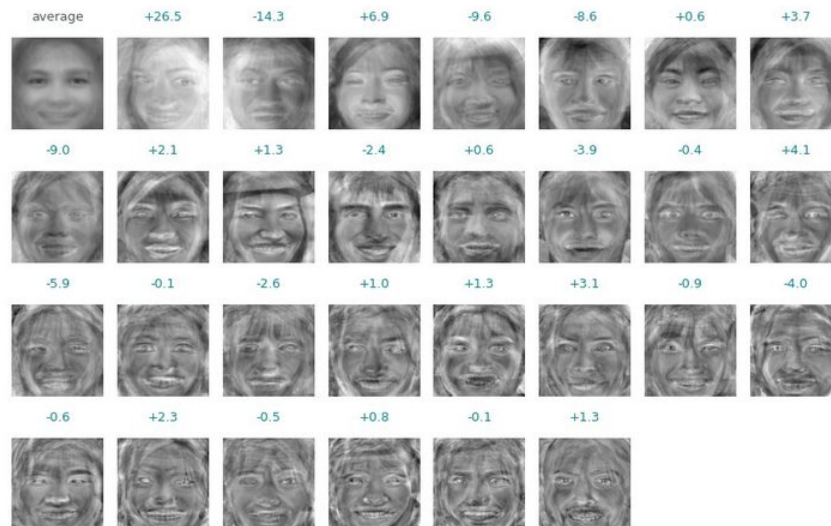


average pixel still 0.8% off

100% captured

Twenty-nine eigenfaces

Exact again. Three people, three lists of twenty-nine numbers, one shared set of eigenfaces.



E1 to E29 in reading order — the teal number over each one is how much of it this face needs



the rebuild is now exact

100 THIRD EXAMPLE

100% captured

The whole sequence, in one table

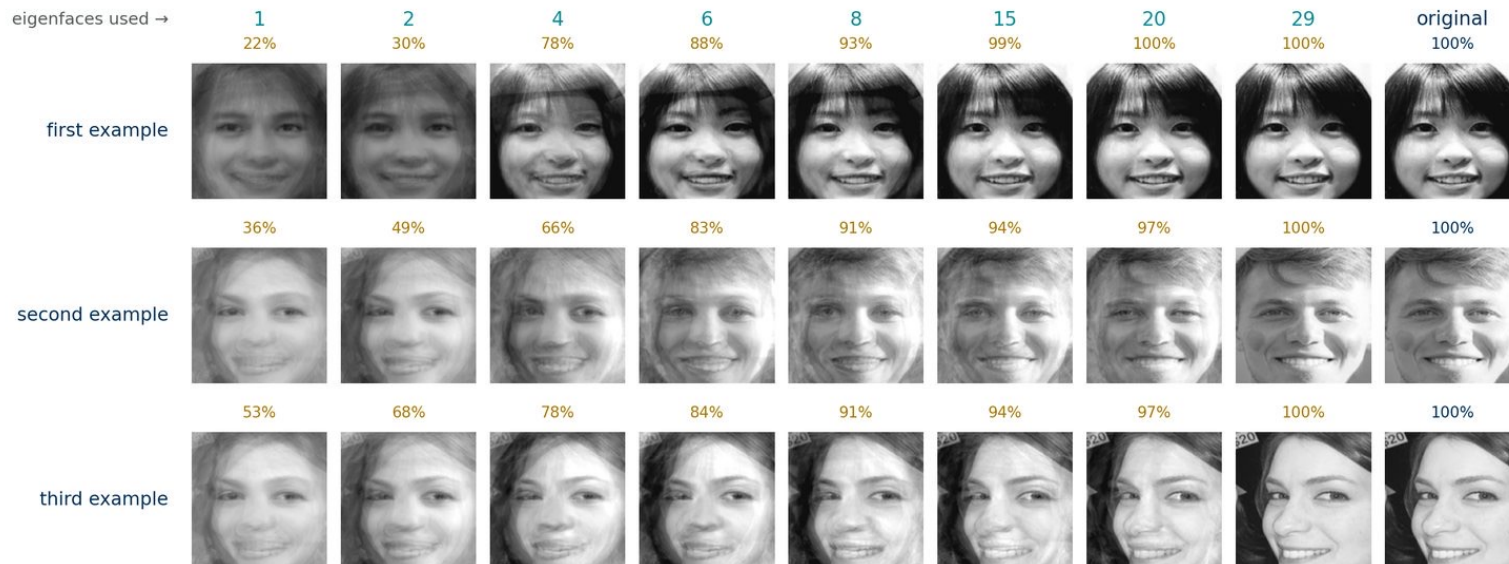


after k	average pixel error	how much of her is captured
1	14.6%	53%
2	12.9%	68%
4	10.3%	78%
6	8.9%	84%
8	6.5%	91%
15	5.2%	94%
20	3.5%	97%
29	0% — exact	100%

The flattest start of the three: E1 alone captures 53% of her. Her head is turned slightly, which the early eigenfaces handle poorly, so the middle of the table is slower to improve.

The same k, three different people

Every frame is built from the same eigenfaces in the same order; only the amounts differ. The number over each frame is how much of that person the rebuild has captured.



the number over each frame is how much of that person the rebuild has captured

So how many eigenfaces are there?

Fair question, and the answer has two halves. The space of 128×128 pictures has 16,384 directions in it, so strictly there are 16,384 eigenvectors. But only twenty-nine of them carry anything at all. The other sixteen thousand describe directions in which all thirty of our photographs sit at exactly the same spot.

16,384

directions the space
has in principle

29

that actually carry
anything at all

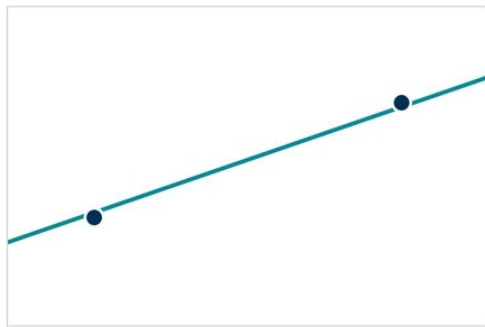
16,355

that are exactly, precisely
empty

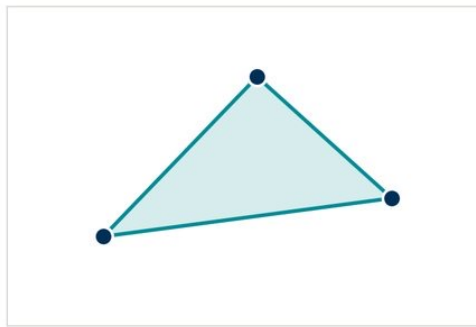
The empty ones are not even unique: any set of directions filling that leftover space works equally well, and the software picks one arbitrarily. Real eigenfaces are pinned down by the data. Those are pinned down by nothing.

Why twenty-nine, and not thirty?

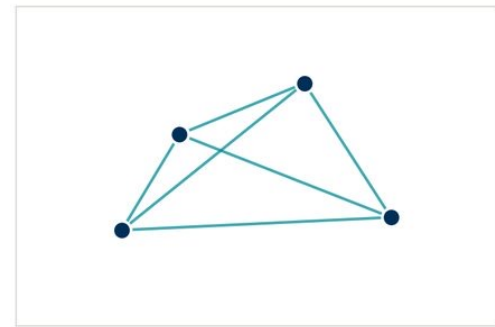
Because n points always fit inside $n - 1$ dimensions. Two points always lie on a line, however big the room. Three always lie on a plane. Thirty photographs always lie inside a twenty-nine-dimensional slice — and subtracting the average face is exactly what uses up that one lost dimension.



2 photographs
always lie on a line
1 dimension



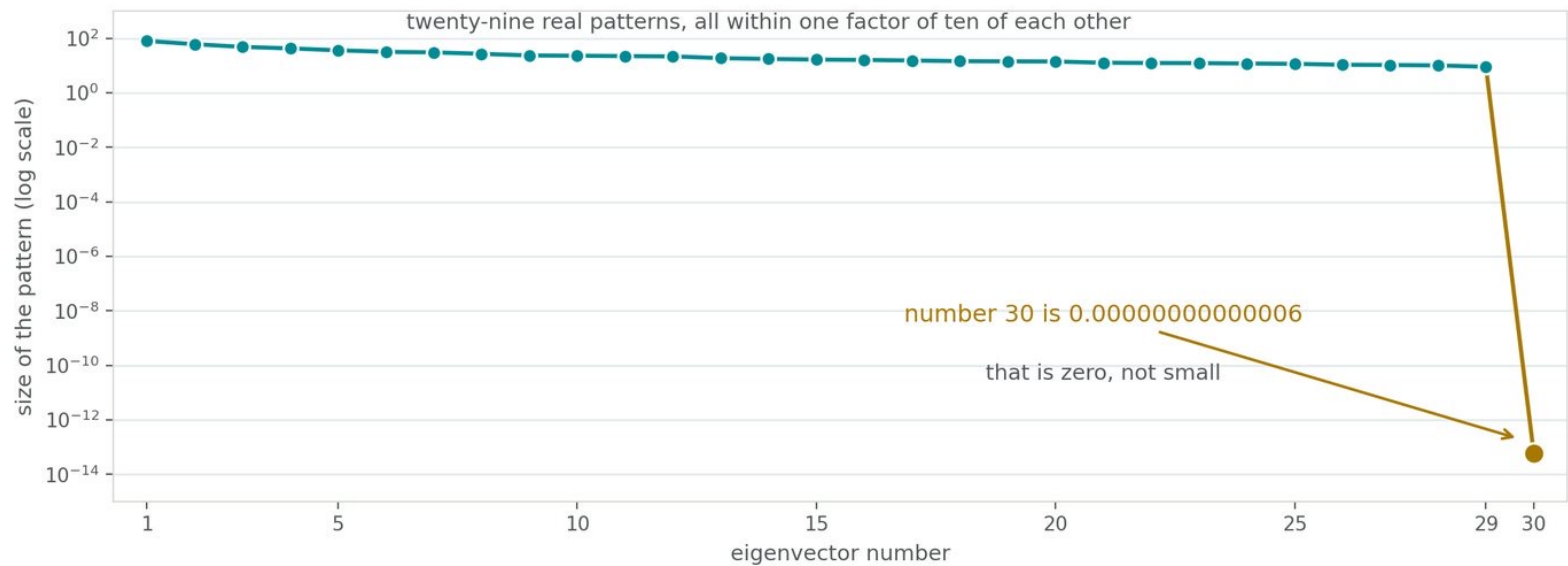
3 photographs
always lie on a plane
2 dimensions



4 photographs
always lie in a 3-D slice
3 dimensions

30 photographs → 29 dimensions → 29 eigenfaces

And the arithmetic says the same thing



then a fall of fifteen orders of magnitude, in a single step

The first twenty-nine patterns range from 80.17 down to 9 — all the same order of magnitude. The thirtieth measures 6.10×10^{-14} , which is floating-point dust: the computer's way of writing zero. There is no thirtieth pattern to find, and no amount of extra pixels would conjure one.

Every eigenface began as a difference

Backing up for a moment: where did those patterns come from? Take a face, subtract the mean face, and you are left with a map of how that person departs from average — brighter here, darker there. PCA never looks at faces at all. It looks at thirty of these difference maps and finds the shapes that keep recurring.



face 03

—



the difference



face 10

—



the difference



face 18

—



the difference

mid-grey = no difference

lighter = brighter than average

darker = darker

The big ones: E1 to E15

Ranked by how much of the variation each one explains. The first four alone account for 60.8% of everything that differs between these thirty people.



E1 27.4%



E2 15.7%



E3 10.0%



E4 7.7%



E5 5.6%



E6 4.3%



E7 3.9%



E8 3.1%



E9 2.3%



E10 2.2%



E11 2.1%



E12 2.0%



E13 1.5%



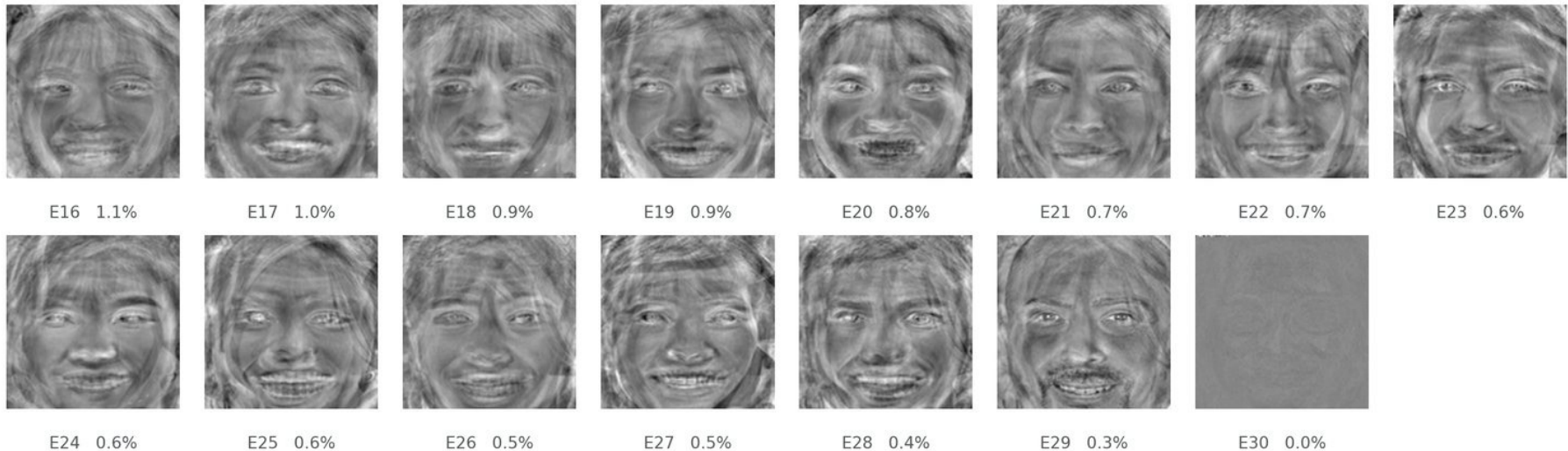
E14 1.3%



E15 1.2%

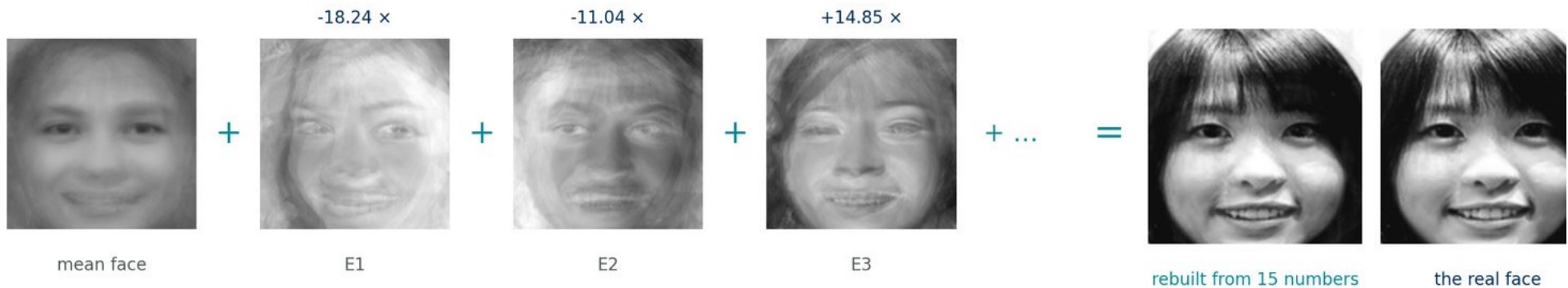
The small ones: E16 to E29

The tail. Each of these explains barely one per cent: fine texture, stray hair, the exact set of a smile. Together E15 to E29 account for just 10.8%. E30 is blank, because with thirty faces there is no thirtieth pattern left to find.



A face is a recipe

Start at the mean face. Add each eigenface in the right amount. The amounts are the only thing that changes from person to person.

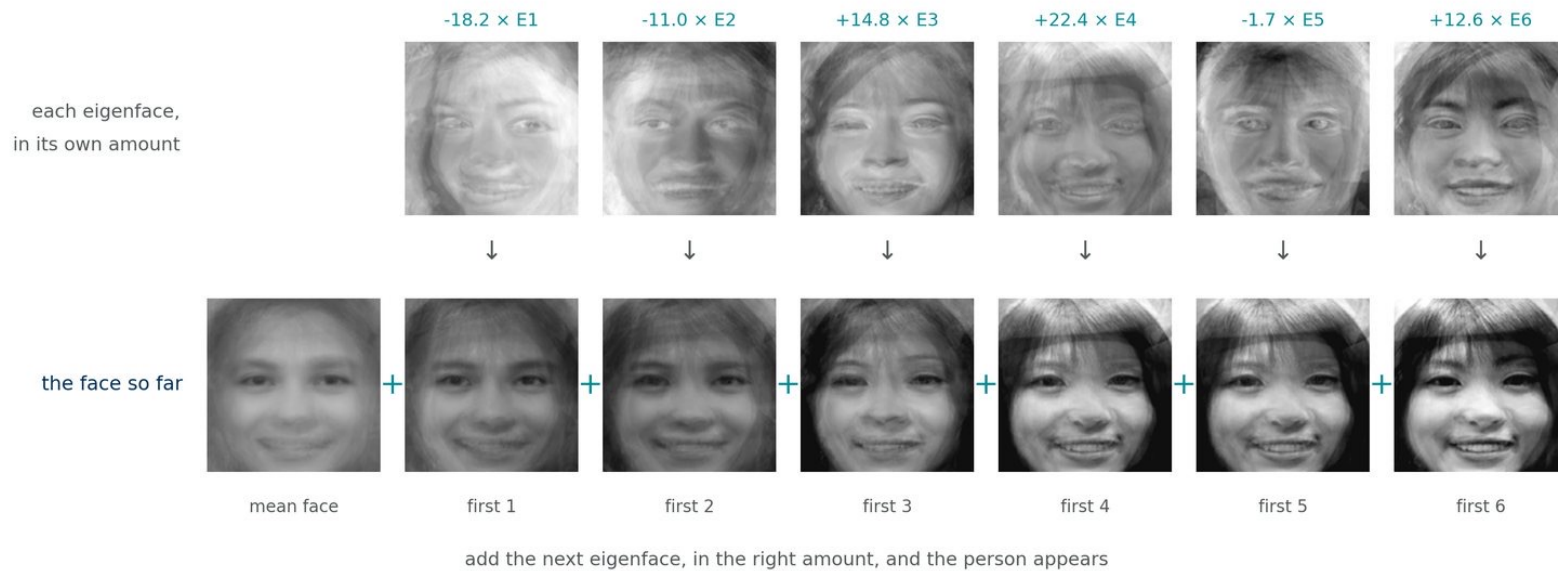


the numbers under the eigenfaces -18.24 , -11.04 , $+14.85$, ... are this person's entire identity

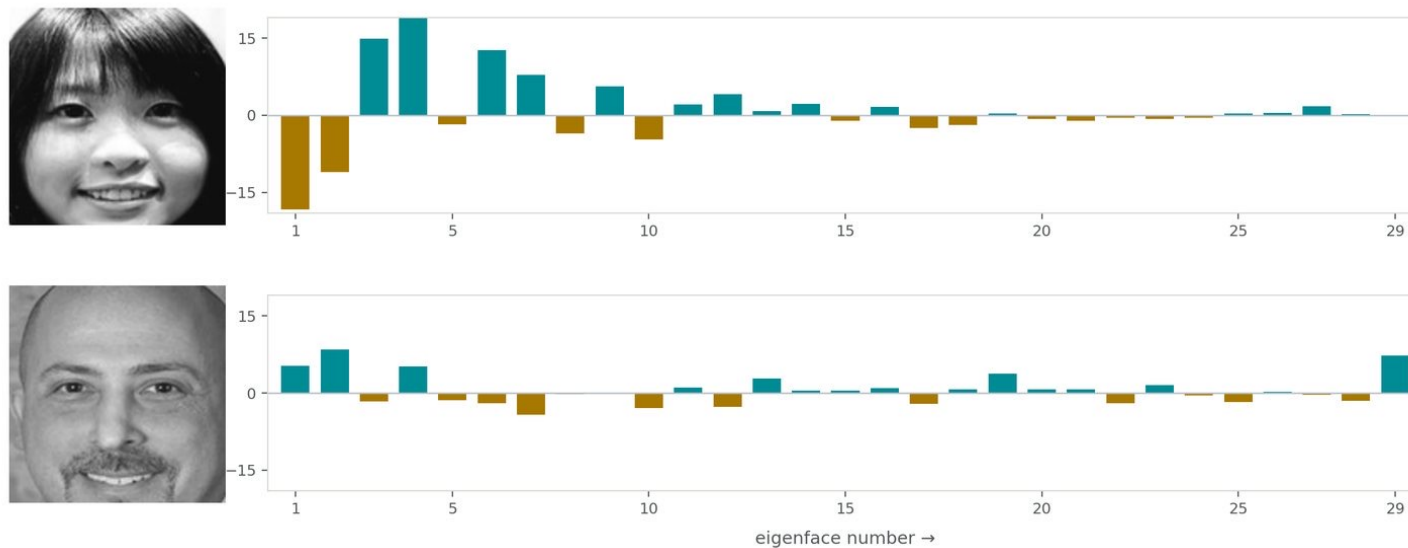
Everyone in the database uses the same E1, E2 and E3. Only the amounts belong to this person — which is exactly why a face compresses so well.

Adding them up, one at a time

Start at the mean face. Add -15.9 of $E1$. Then -8.5 of $E2$. Then $+7.7$ of $E3$. Every step is the same move: take an eigenface, scale it, add it on. Nothing else happens.



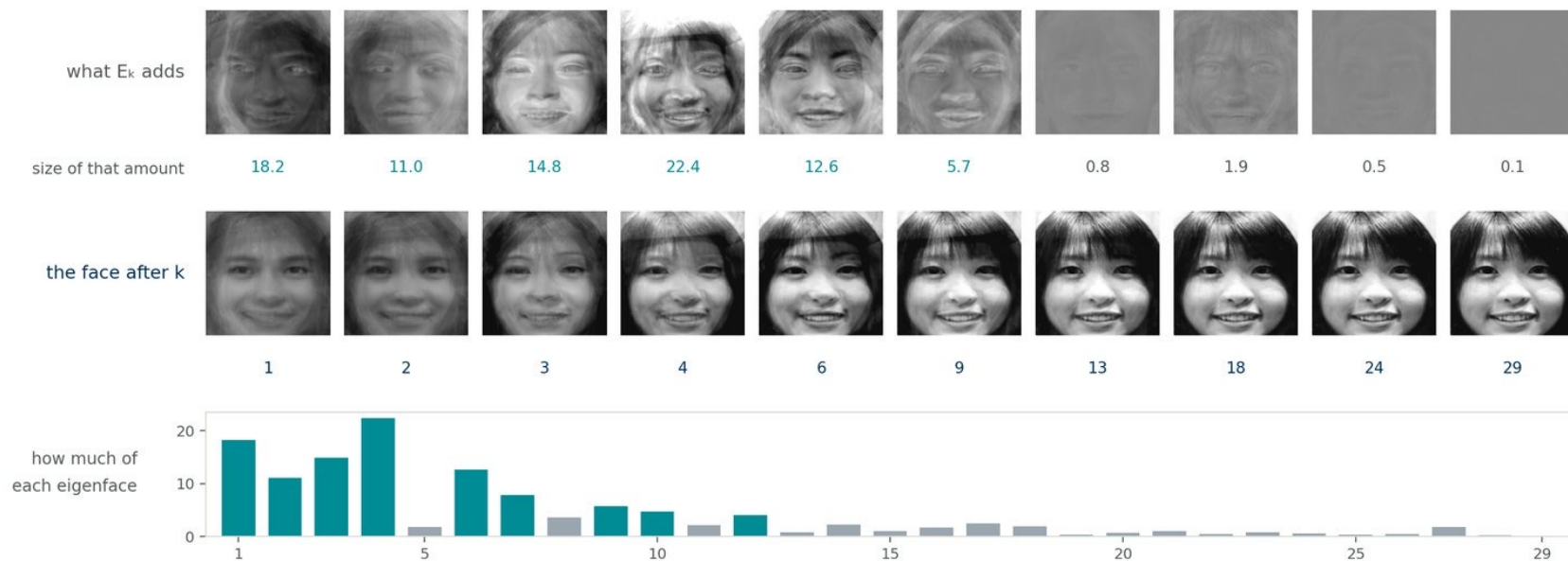
Which means a face is a list of 29 numbers



the same 29 eigenfaces both times — only the 29 amounts change

Same twenty-nine eigenfaces for both people. Everything that makes one of them not the other is in the heights of those bars — and a bar chart is a much smaller thing to store than a photograph.

Most of the eigenfaces barely matter



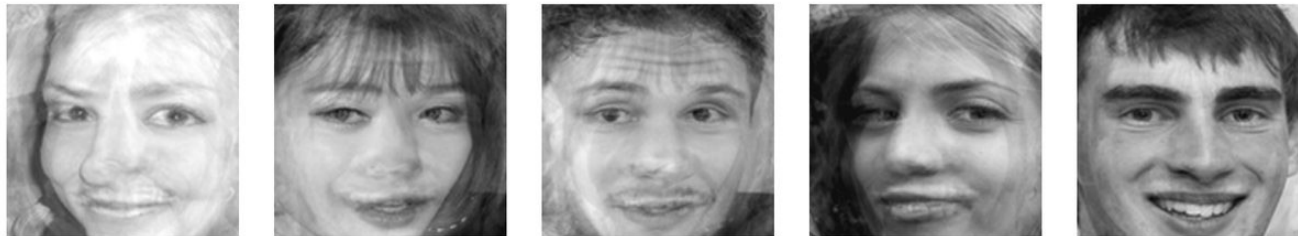
The top row shows what each eigenface actually contributes to this face, all drawn at the same scale — by E18 there is nothing left to add. The first eight carry 93% of her; the remaining twenty-one carry 7% between them. Throwing the small ones away is exactly what compression means.

Everyone rebuilt from the same fifteen patterns

the real photo



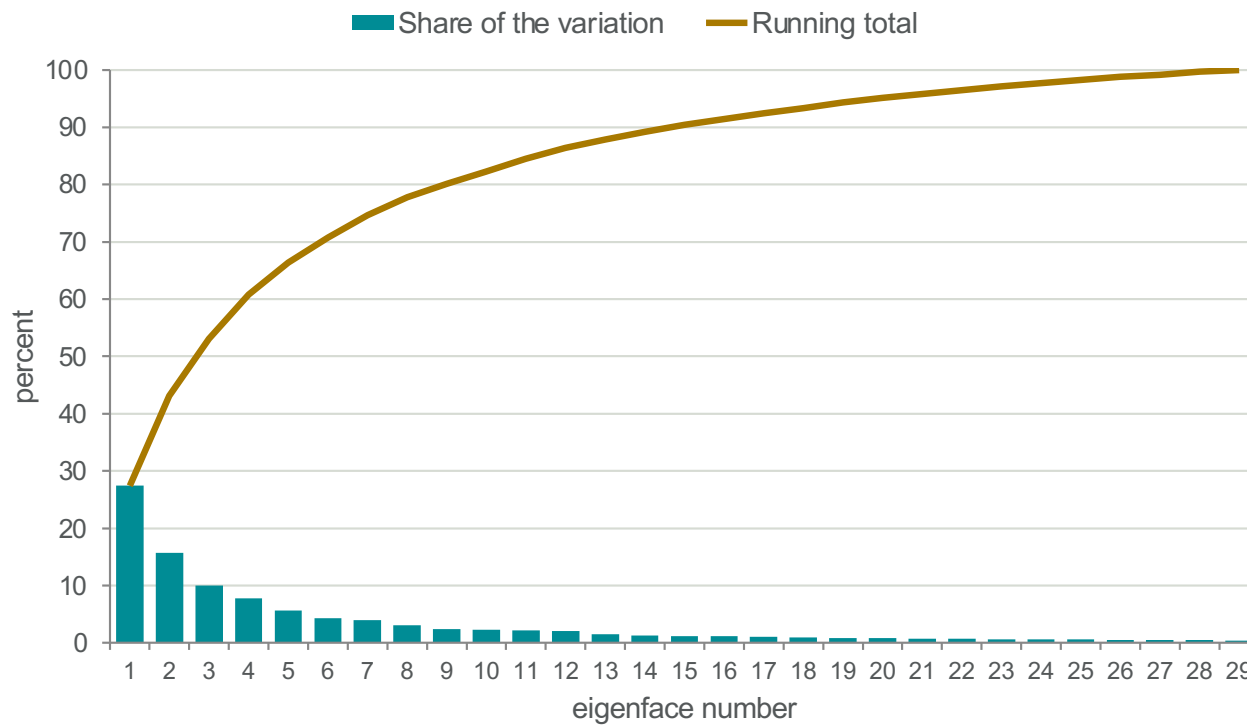
rebuilt from
15 numbers



the same fifteen eigenfaces, in different amounts

16,384 numbers → 15 numbers per person: a thousand-fold saving. Honest footnote: the fifteen eigenfaces and the mean face have to be stored too — but only once, however many people you have.

Where the information actually sits



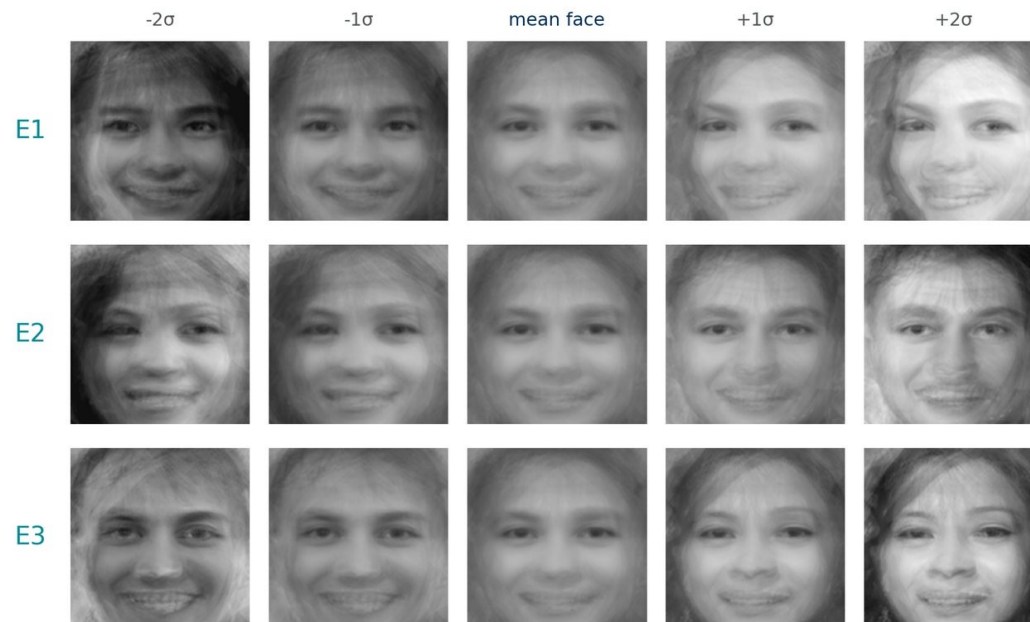
E1 alone
27%

first 5
66%

first 8
78%

first 15
90%

Turn the knobs yourself



each row turns one knob — nobody told PCA what these knobs should mean

- Start at the mean face and dial each eigenface up or down.
- E1 is exposure — how brightly the photo was lit.
- E2 decides whether the face is lighter than its background, or darker.
- E3 controls how much dark hair frames the face.
- Nobody labelled these knobs. PCA found them by looking at thirty examples — which is what makes it learning rather than programming.

Feed it something that is not a face



a cat



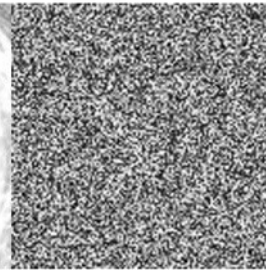
rebuilt with all 29



a car



rebuilt with all 29



static



rebuilt with all 29

These eigenfaces were learned from thirty human faces, so a face is the only thing they can build. A cat, a car and pure static all come back wearing the same ghostly expression — even with all 29 components switched on. PCA only knows what it was shown.

Writing it down properly

Everything you have just watched, in five slides of notation. Nothing new happens here — this is the same procedure, written so it can be checked.

$$x_1 \dots x_{30} \in \mathbb{R}^{16,384}$$

each photograph, unrolled into one long column of pixel values

$$\mu = (1/N) \sum x_i$$

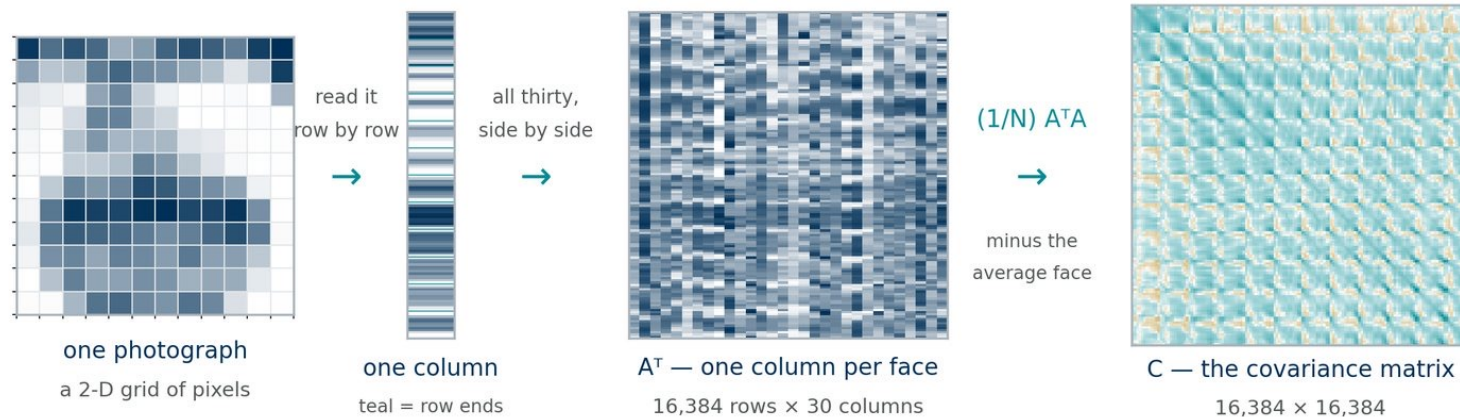
the average face

$$a_i = x_i - \mu \quad A = [a_1 \dots a_{30}]^T \quad (30 \times 16,384)$$

the difference maps, stacked into one matrix. Note $\sum a_i = 0$ — that constraint is the dimension we lose.

From a 2-D picture to a matrix

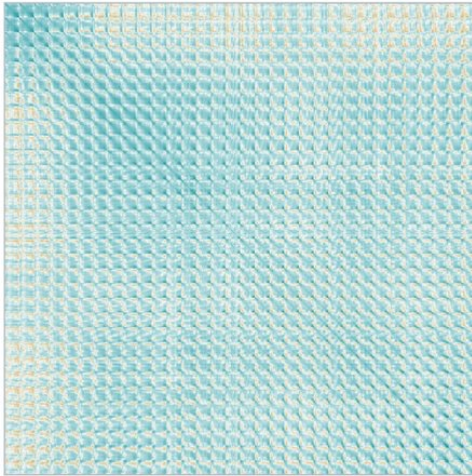
The step everything else rests on. A photograph is a 2-D grid, but linear algebra wants a column of numbers — so read the grid row by row and stand it up. Do that for all thirty, set the columns side by side, subtract the average, and multiply the block by its own transpose.



drawn with 12×12 pictures so the wiring stays visible — the real ones are 128×128 , so each column is 16,384 tall

The covariance matrix

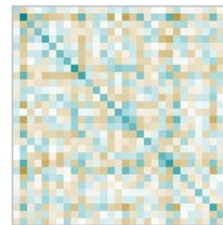
$$C = (1/N) A^T A \quad C_{jk} = \text{how pixel } j \text{ and pixel } k \text{ vary together}$$



the covariance matrix C

one row and one column per pixel

drawn from 32 × 32 versions of the photographs, so that C is $1,024 \times 1,024$ instead of $16,384 \times 16,384$ — the real one would not fit on a screen



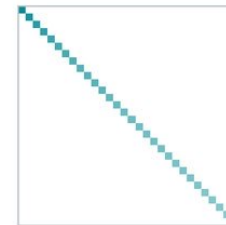
the Gram matrix G

one row per face

same eigenvalues



546× smaller



its eigenvalues λ

29 non-zero, 1 zero

The eigenvalue problem

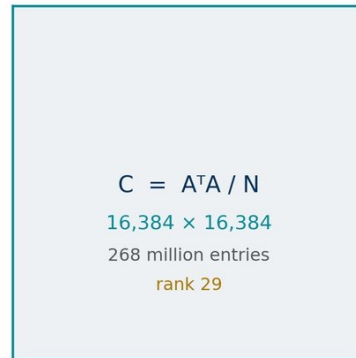
$$C u_j = \lambda_j u_j$$

$$\|u_j\| = 1, \quad u_j^T u_k = 0 \text{ for } j \neq k, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{29} > 0$$

- Each eigenvector u_j is one eigenface: a direction in the space of pictures.
- Its eigenvalue λ_j is the variance of the thirty faces along that direction — literally how spread out they are that way.
- Sorting by λ is what makes E1 the most important pattern rather than an arbitrary one.
- $\lambda_j / \sum \lambda$ is the percentage printed under every eigenface in this deck.

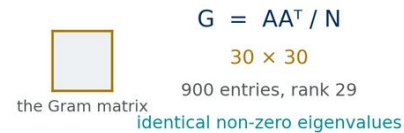
The trick nobody skips

You never build C . It has 268 million entries and a rank of 29 — almost all of it is structurally empty. Work with the small matrix instead and map the answer back.



the covariance matrix

(drawn 30× too large — at true scale it is a speck)



$$G v_j = \lambda_j v_j \quad \rightarrow \quad u_j = A^T v_j / \|A^T v_j\| \quad \text{— same } \lambda, \text{ same eigenfaces}$$

$A A^T$ and $A^T A$ always share their non-zero eigenvalues, and $A^T A$ can have no more of them than A has rows. That single fact is both the computational shortcut and the reason the answer is 29.

Projection, reconstruction, error

$$c_j = u_j^T (x - \mu)$$

$$\hat{x} = \mu + \sum_{j=1}^k c_j u_j$$

$$\text{average over the set of } \|x - \hat{x}\|^2 = \lambda_{k+1} + \lambda_{k+2} + \dots + \lambda_{29}$$

- The first line is the whole deck: measure the amounts, then rebuild from them.
- The error you are left with is exactly the eigenvalues you threw away — which is why the variance chart doubles as an error budget.
- No other k-dimensional flat subspace does better. PCA is provably the best linear choice, not merely a reasonable one.
- Take all 29 and the error is zero, because nothing has been discarded.

Three things to take away

1

Data has a grain.

Measurements that look independent usually are not. PCA finds the few directions things really vary along.

2

A face is a short list.

Mean face plus about fifteen amounts. The patterns are shared; only the amounts are yours.

3

It only knows what it saw.

Thirty faces buy you 29 knobs, and anything that is not a face comes back as one.

Same method, no faces required: sorting genes, compressing sensor data, spotting a wobble in a machine before it breaks.